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MILITARY SPECIFICATION

COMMUNICATIONS CONTROL MONITOR GROUP AIR TRAFFIC CONTROL AN/FSA-58(V)

1.0 SCOPE

1.1 Scope. This specification covers the design, construction, and test of a fully modularized control monitor group intended for air traffic control and monitor operations. The AN/FSA-58(V), hereinafter called the equipment, is similar in operation and configuration to the AN/FSA-58(XN-1)(V); however, additional features and solid state switching are required in the AN/FSA-58(V).

2.0 APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

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MIL-P-116	Preservation-packaging, Methods Of
MIL-L-3661	Lampholder, Indicator Lights, Indicator-Light Housings, And Indicator-Light Lenses, General Specification For
MIL-L-6363	Lamp, Incandescent, Aviation Service, General Requirements For
MIL-L-12606	Loudspeaker, Permanent Magnet (Unencased, 4-inch Diameter Cone, 2-Watt, Gunblast, and Immersion-Resistant), Type LS-445/U
MIL-E-16400	Electronic Interior Communication And Navigation Equipment, Naval Ship And Shore, General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-F-19207	Fuseholder, Extractor Post Type, Blown Fuse Indicating And Non-indicating, General Specification For

STANDARDS

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MIL-STD-108	Definitions Of And Basic Requirements For Enclosure For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-188-346	Military Communication System Technical Standards
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-454	Standard General REquirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (code 5043) Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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HANDBOOKS

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MIL-HDBK-241 Design Guide For EMI Reduction In Power Supplies

(Copies of specifications, standards, handbooks, drawings, and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UL-492 Standard For Radio And Television Receiving Appliances (Underwriters Laboratories, Inc.)

(Copies may be obtained from Underwriter's Laboratories Incorporated, 1285 Walt Whitman Road, Melville, Long Island, N. Y. 11746.)

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3.0 REQUIREMENTS

3.1 First article sample. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.4).

3.2 General. Except as hereinafter specified, the equipment shall be in accordance with the requirements of MIL-E-16400. Particular consideration shall be given to sections specified in MIL-E-16400 governing cable harness, wire dress, workmanship, soldering techniques, and plug-in design techniques.

3.2.1 Temperature. The equipment shall conform to the temperature range requirements of MIL-E-16400 for Class 4 equipment.

3.2.2 Primary power line voltage and frequency. The equipment shall operate within the requirements specified herein when connected to a source of 115 volts, plus or minus 10 percent, 50 to 60 Hertz (Hz), plus or minus five percent, single-phase electrical power.

3.2.3 Ruggedness. The equipment shall be sufficiently rugged, electrically and mechanically, to withstand severe and prolonged use, shipping, and maintenance handling, and the shocks and vibration accompanying such use.

3.2.4 Quantitative reliability requirement. The specified mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) of this equipment shall be 600 hours for the basic equipment as specified in 3.3.1 through 3.3.1.5.

3.2.5 Quantitative maintainability requirements. The equipment shall have a specified median-corrective-maintenance-time (M_{ct}) of not more than 0.25 hours when corrective maintenance is accomlished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, p.c. boards) and chassis-mounted components (this includes electronic, electrical, electro-mechanical, and mechanical parts). The equipment corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions.

3.2.6 Maintenance design.

3.2.6.1 Circuit test points. Circuit test point accesses shall be provided and shall comply with the criteria outlined in MIL-STD-415. The following types of tests shall be considered for determining the test points:

a. Signal quality
b. Circuit performance
c. Calibration
d. System alignment
e. Installation and check-out
f. Planned maintenance systems tests
g. Line replaceable unit (LRU) fault location
h. Shop replaceable part (SRP) fault isolation

Test points for calibration shall be provided external to unit under test (UUT), except where it would affect the calibration. Test points shall be provided with proper interface fittings and matching circuits to connect the selected support test equipment.

3.2.7 Locking devices. Operator control locking devices are not required.

3.2.8 Time meters. Time meters are not required.

3.2.9 Intra-unit wiring. All conductors shall be stranded conductors, No. 22 gauge with 600-volt insulation except where otherwise specified herein. Patch cords shall be No. 22 gauge made up of 19 strands of No. 35 gauge tinned copper.

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3.2.10 Power connections. Wherever 115 VAC 60 Hz power connections are made, equipment power connectors shall have pins designated A, B, and C. Pin A shall be connected to the low (return) side and Pin C to the high (hot) side of the power source. Pin B shall be connected to the internal chassis by means of a conductor at least equal in size to one of the conductors supplying power.

3.2.11 Degree of enclosure. Degree of enclosure shall be dustproof in accordance with MIL-STD-108.

3.2.12 Human factors engineering. The contractor shall comply with the Human Factors Design Criteria of MIL-STD-1472 and the requirements specified herein. If off-the-shelf equipment is used, it shall be selected as required by the Standardization paragraph of MIL-STD-1472.

3.2.12.1 Controls, indicators, and panel layouts. Design of operator and maintainer panel layouts and controls shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, Indicators and Panel Layout paragraph of MIL-E-16400, except that color coding shall be in accordance with MIL-STD-1472 and as otherwise specified herein.

3.2.12.2 Indicator and illumination lamps. Incandescent lamps, Type T-1-3/4 miniature flange base, 28 Volts, 0.04 Amps, in accordance with MIL-L-6363 for 20,000 hour rated life, shall be incorporated throughout except where otherwise specified herein.

3.2.13 Electromagnetic compatibility (EMC) characteristics. The equipment shall be self-compatible and shall conform to the requirements for Class 1C of MIL-STD-461, as follows: CE01, CE03, RE02, and RS01. In lieu of the remaining Class 1C requirements of MIL-STD-461, the contractor shall demonstrate that transients from relay activation or deactivation and transmitter keying shall not cause interference in either radio receivers or transmitters. MIL-HDBK-241 may be used as a design guide for EMI reduction in power supplies.

3.2.14 Single voltage direct current (VDC) power operation. All components of the AN/FSA-58(V) except AN/FSA-32 and AN/FSA-47(V) shall operate from a single power supply voltage of a nominal +25 VDC. These components shall operate in accordance with requirements specified herein from 23 to 28 VDC and shall perform all operational functions from 21 to 23 VDC and 28 to 30 VDC without damage to the equipment at a reduction in performance.

3.2.14.1 Floating ground option. The equipment shall be designed to be installed with either a hard ground or floating ground by strapping means for all control subgroups.

3.2.15 Equipment hum and noise. The hum and noise level shall be minus 60 dBm in the output of each amplifier with rectifier/charger and battery operation.

3.2.16 Standard electronic modules (SEM). Standard electronic modules are not required.

3.2.17 Solid state switching. The AN/FSA-58(V) shall have solid state switching for interphone, radiophone, transmitter selection, and control functions.

3.3 Design and construction. The major components of the equipment shall be:

a. Operator control subgroup
b. Supervisor control subgroup
c. AN/FSA-32
d. AN/FSA-47(V)
e. Control central subgroup
f. Support and diagnostic subgroup

3.3.1 Basic equipment. The basic AN/FSA-58(V) configuration, to provide control, monitor, and switching of radiophone, private interphone, and landline (telephone) communications at operator and supervisor positions for air traffic control operations shall be as specified in 3.3.1.1 through 3.3.1.5.

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3.3.1.1 Operator control subgroups. Two operator control subgroups shall be required for installation in the radar air traffic control center (RATCC) or remote location, each consisting of:

a. One radiophone switch unit (RSU) with radiophone switch modules (RSM) for selection and control of five radiophone channels
b. One private interphone/monitor switch unit (PIMSU) with private interphone switch modules (PISM) for selection and control of nine interphone channels and six position monitor switch (PMS) modules for six monitor channels
c. One speaker amplifier unit (SAU)
d. One console amplifier unit (CAU)
e. One foot switch, SA-585/FSA-17
f. One jackbox J-( )/FSA-17, with cable connectors
g. Two headsets, H-170/U with noise canceling microphone, Roanwell Corporation Model RN-1-C, or equivalent
h. Cable terminal block connector harnesses, quantity as required
i. Add-on cable terminal block connector harnesses, quantity as required
j. Mounting bracket kits, quantity as required
k. One landline (telephone) dial unit 58 of OA-7621(V)/FSA-52(V) packaged in accordance with 3.3.3.14
l. One landline (telephone) selector and control unit 59 of OA-7621(V)/FSA-52(V) packaged in accordance with 3.3.3.15 for 15 channels

3.3.1.2 Operator control subgroup (control tower). Three operator control subgroups shall be for installation in the air traffic control tower, each consisting of the items listed in 3.3.1.1 and the following:

a. One AN/FSA-32 switching panel SB-1281/FSA-32 (see 3.3.5.10)
b. One control tower operator jackbox
c. One noise canceling hand-held microphone with transmitter key switch
d. Eight speakers (Type LS 445/U)
e. One modular unit jackbox

3.3.1.3 Supervisor control subgroup. One supervisor control subgroup shall be for installation in the radar air traffic control center, consisting of the items listed in 3.3.1.1 and the following:

a. One each AN/FSA-47(V) remote panel SB-1996/FSA-47 and dimmer control C-4792/FSA-47 (for five channels and dimmer control) with associated cabinet CY-4042/FSA-47
b. Switches for control of all radiophone channels on an override basis
c. Status switch units (SSU) with status switch modules (SSM) for complete equipment status display
d. Cable terminal block connector harness, quantity as required
e. Mounting bracket kits for additional modules, quantity as required
f. One supervisor special control module (SSCM)

3.3.1.4 Control central subgroup. One control central subgroup, for installation in the terminal equipment room, shall consist of:

a. One transmitter-operator select/control unit for 10 radiophone channels
b. One receiver amplifier indicator unit for 10 radiophone channels with 10 amplifier modules
c. Radiophone program unit for five operator control subgroups for programming 10 selector switches from a selection of 20 channels
d. One monitor panel unit
e. One private interphone program unit
f. One fuse/power distribution panel unit
g. One AN/FSA-47(V) local panel SB-1995/FSA-47 (five channels), one power supply panel PP-3824/FSA-47, with associated cabinet CY-4040/FSA-47

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h. One set of AN/FSA-32 components consisting of the following items specified in 3.3.5.8
   1. Three mixer amplifiers AM-2984/FSA-32
   2. Eight amplifiers, audio frequency AM-2983/FSA-32
   3. One power supply PP-2947/FSA-32
   4. One cabinet, electrical equipment CY-3129/FSA-32
i. One landline interface panel
j. One power supply unit
k. Program patch cords, radiophone, quantity as required
l. Program patch cords, interphone, quantity as required
m. Program patch cords, position monitor, quantity as required
n. One monitor program unit

3.3.1.5 Support and diagnostic subgroup. The support and diagnostic subgroup, for use in the installation and maintenance of the remaining components of the AN/FSA-58(V), consists of as a minimum:

a. One special test set for the AN/FSA-58(V)
b. One special audio test set TS-1505A/FSA-17
c. One test set electronic, TS-1664/FSA-32
d. Two sets of riveting and deriveting tools
e. Rivets for full installation plus 100 percent spares
f. Two sets of wire wrap and unwrap tools
g. Two sets of extender cards

3.3.2 Variable capability. The equipment shall be variable as specified in 3.3.2.1 through 3.3.2.3.8.

3.3.2.1 Operator control subgroup.

3.3.2.1.1 Internal variation. Operator control subgroup shall be variable by addition or deletion of radiophone and interphone selector units. Each radiophone unit shall consist of five switch modules for control of five channels programmed from the total available to the equipment. Each interphone unit shall consist of five switch modules for control of 15 interphone or monitor channels in any combination of switch modules.

3.3.2.1.2 Quantity. The AN/FSA-58(V) shall function as specified herein with one to at least sixty operator control subgroups and one supervisor control subgroup connected as required through the control central subgroup.

3.3.2.2 Supervisor control subgroup. Supervisor control subgroup shall be variable by the addition or deletion of radiophone and interphone units. The supervisor control subgroup shall have control of all radiophone, interphone, and monitor channels available in the system. Status display units shall be added to provide the display and special functions required for all radiophone channels, and operator control subgroups.

3.3.2.3 Control central subgroup. Control central subgroup shall be variable as specified in 3.3.2.3.1 through 3.3.2.3.8.

3.3.2.3.1 Radiophone channels. Radiophone channels shall be variable by single channel modules up to 10 modules for each receiver and transmitter control unit and up to six units each per system.

3.3.2.3.2 Radiophone programming. Radiophone programming shall be variable by one program panel unit for five operator subgroups with any 10 channels to each subgroup from a selection of 20 radiophone channels for each subgroup.

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3.3.2.3.3 Interphone programming. Interphone programming shall be variable by one interphone program panel unit with interphone capability for eight control subgroups with 15 channels each subgroup.

3.3.2.3.4 Monitor programming. Monitor programming shall be variable by one unit which provides programming to 12 operator subgroups to monitor six operator control subgroups on an individual basis.

3.3.2.3.5 Transmitter control unit. One transmitter control unit for 10 transmitters.

3.3.2.3.6 Receiver amplifier unit. One receiver amplifier unit for 10 radiophone channels.

3.3.2.3.7 Power supply unit. One power supply unit for six positions and 20 radiophone channels (power supply units vary from 12 to 100 amperes).

3.3.2.3.8 Control transfer group. One control transfer group (AN/FSA-47(V)) in increments of five channels each with up to five increments per power supply.

3.3.3 Operator control subgroup. Each operator control subgroup shall have all the necessary switches, controls, circuitry, indicators, and elements specified in 3.3.1.1 and 3.3.1.2 to provide communications for air traffic control and monitor operations.

3.3.3.1 Control panel configuration. The control panel shall be made up of any number of radiophone and interphone switch units, one speaker amplifier unit, one landline selector unit, one dial unit, one console amplifier unit, jackbox unit, one or more blank units, and mounting flanges riveted together to mount in both 19- and 24-inch panel spaces. The units shall be capable of being mounted in both vertical and horizontal positions. Each unit shall be no more than 133.4 mm (5.25 inches) in width, 88.9 mm (3.50 inches) in height, and less than 101.6 mm (four-inches) deep overall with connectors. As specified herein, the 88.9 mm (3.50-inch) dimension in the vertical plane and 133.4 mm (5.25-inch) dimension in the horizontal plane shall be considered horizontal mounting. Unless otherwise specified herein the units and plug-in modules shall be as described in the horizontal mounting position. The unit cases shall be fabricated in accordance with FIGURES 1 and 2, with a depth of no more than 101.6 mm (four inches) overall with connectors unless otherwise approved by the procuring activity. The unit case shall be interchangeable for all type units. Each unit case shall contain receptacles for accepting plug-in module(s). Edge panel lighting shall be provided for horizontal and vertical mounting.

3.3.3.1.1 Control panel unit. Each unit shall consist of a case and plug-in module(s). A blank panel unit shall consist of a case and blank panel.

3.3.3.1.2 Switch module non-interchangeability. Amplifier modules, landline, radiophone, interphone, and supervisor status switch modules shall be keyed to ensure connection in only the designated cases.

3.3.3.2 Single plane. All indicators, manual controls, and meters necessary for the operation of the equipment, except main power ON-OFF switch, shall be located in a single plane in the operator control subgroup unless otherwise specified.

3.3.3.3 Control circuitry. Circuitry shall be provided in each operator control subgroup to provide the control features specified in 3.3.3.3.1 through 3.3.3.3.14.

3.3.3.3.1 Radiophone monitoring. Each operator and supervisor control subgroup shall be provided the capability to monitor all communications on each radiophone channel programmed to the subgroup. The operator shall be able to select and monitor any number of radiophone channels in any combination. Speaker and headset monitoring shall be provided for each channel in any combination of the channels available at the operator control subgroup.

3.3.3.3.1.1 True sidetone. True sidetone from the radiophone receiver circuit shall be provided in the headphone of the keying subgroup.

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3.3.3.3.2 Simultaneous transmission. Each operator control subgroup shall enable its operator to select and modulate simultaneously any combination of programmed radio transmitters, landline channels, and any selected combination of private interphone circuits.

3.3.3.3.2.1 Landline channel conference. Circuitry of the AN/FSA-58(V) shall, without any modification of the OA 7621(V)/FSA-52(V), interface with the conference capabilities of the Landline Selector Group OA-7621(V)/FSA-52(V).

3.3.3.3.3 Private interphone. Each operator control subgroup shall enable its operator to select and call other operator(s) and the supervisor by normal call, override, and Instantaneous Interphone Override Response (IIOR).

3.3.3.3.4 Private interphone conference. Each operator control subgroup shall enable its operator to initiate an interphone conference with any combination of other operators and the supervisor without disturbing any other control subgroup and the operators.

3.3.3.3.5 Speaker override call. Each operator control subgroup shall enable the operator to receive an interphone call on an override basis in the position speaker. On answering, the interphone call shall be automatically transferred to headset operation.

3.3.3.3.6 Operator subgroup monitoring. Each operator control subgroup shall enable its operator to select and monitor all communications at any other operator control subgroup and supervisor control subgroup programmed to the selecting operator control subgroup.

3.3.3.3.7 Landline interface. Three methods of landline connections shall be provided.

3.3.3.3.7.1 Radiophone circuits for control of landlines. Control of voice signalling (in and out) landline channels shall be provided through the radiophone control circuitry with transmission when channel is selected for transmit.

3.3.3.3.7.2 Jackbox control transfer switch. Operation of a switch in the subgroup jackbox shall transfer the headset to leased landline equipment and the radiophone and interphone channels monitored in the headset shall automatically transfer to the subgroup's speaker when a headset is plugged into the RADIO TELCO jacks. A blue lamp on the jackbox shall automatically indicate the headset transfer to the leased landline equipment.

3.3.3.3.7.3 Integrated operation with OA-7621(V)/FSA-52(V). Wiring and terminals shall be provided in the console amplifier case for OA-7621(V)/FSA-52(V) interface.

3.3.3.3.8 Sidetone level. The operator's sidetone shall be present only in the operator's headset with any combination of radiophone, interphone, and landline channels selected. The operator's sidetone shall be 12-dB lower than an incoming signal level.

3.3.3.3.9 Simultaneous transmission and monitor. Simultaneous transmission on any number of radiophone, interphone, and landline channels shall be provided in any combination within the limits of channels programmed to the operator control subgroups. Radiophone and interphone channels shall be selected at the control subgroup for its speaker or headset monitor in any combination. Radiophone channels shall be pre-programmed in the AN/FSA-32 overhead speakers for control tower operator monitoring. During transmission, radiophone, interphone, and landline channels shall be monitored in the control subgroup and overhead speakers except for those channels being keyed. The keyed channels shall automatically transfer to headphone for sidetone monitor with other channels selected for headphone monitor.

3.3.3.3.10 Operator control panel illumination levels. Illumination of all switches, indicators, lamps, meters, and edge-lighted panels shall be provided for operation in a bright light and dark environment. The illumination shall be continuously variable from a dim light for the extreme dark night to full brilliance for the intense bright sun light environment of the control tower cab. Illumination of all control panels and switches at an operator subgroup shall be varied by a single control.

3.3.3.3.10.1 Lamp indication voltage. The voltage required for full brilliance indication shall be no less than 23.5 VDC across the associated lamp with 25.0 VDC input to the position.

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3.3.3.3.11 Operator audio level controls. There shall be a variable level control for the headphone amplifier and one variable control for the control subgroup speaker amplifier.

3.3.3.3.12 Technician audio level controls. There shall be one multi-turn, card-mounted potentiometer audio control for infrequent use in the microphone amplifier. There shall be one variable control for infrequent use to pre-set the attenuation level for the control subgroup speaker mute switch.

3.3.3.3.13 Equipment audio levels. The equipment audio levels shall be on a fixed-level basis for zero dBm input into the radio receive circuit and zero dBm into the transmitter audio circuit.

3.3.3.3.14 AN/FSA-32 interface. Each operator control subgroup shall provide for automatic muting in the overhead speaker of a keyed radiophone channel. A control panel shall be provided for each of three tower operator control subgroups for muting of overhead speakers individually and one switch for muting all speakers simultaneously. The AN/FSA-32 channel programming shall correspond to the channel switch programming of the three tower operator control subgroups which shall be identically programmed.

3.3.3.4 Control switches. There shall be one switch module with three switches for each radiophone channel, one switch module with three switches for three interphone channels, one switch for subgroup speaker mute, one switch in the jackbox for transfer to leased landline equipment with automatic transfer of headset monitoring to speaker, one foot bar switch for keying function, jackbox switch for automatic transfer of headset monitor to speaker upon removal of headset plugs from jackbox, one transmitter key switch in a hand-held microphone, and one in-cord headset switch for microphone. Control circuitry shall be provided to permit simultaneous operation of switches for transmission and monitoring of channels in any combination without degradation to equipment operation. Each position of a switch shall be identified by a physical displacement of the switch button or lever arm. Panel markings shall identify the switch position function for the lever switches.

3.3.3.4.1 Pushbutton selector switch modules (SSM). Pushbutton switches with lights shall be provided for selecting radiophone, interphone, position monitor, telephone channels, supervisor special control, and equipment status displays in the supervisor control subgroup. Three switches shall be mounted on one plug-in card (switch module). The molded switch frame of unit 59A1 of the OA-7621(V)/FSA-52(V) shall be used for all six switches. Two lamps shall be provided in each switch. Five switch modules plugged into a console shell (case) shall make up one selector switch unit. The switches shall be push ON and push OFF except where otherwise specified herein.

3.3.3.4.2 Radiophone switch modules (RSM). Each RSM shall contain three pushbutton switch lights for control of one radiophone channel.

3.3.3.4.2.1 RSM receiver monitor switch. The lower pushbutton in the out position shall indicate the channel is not monitored at the operator control subgroup and the switch light shall be out. The pushbutton in a depressed position shall indicate that the channel is selected for monitor and an amber light shall be on at nominal half brilliance. Full brilliance (3.3.3.3.10.1) shall indicate the presence of a signal regardless of the switch position.

3.3.3.4.2.2 RSM headset-speaker monitor switch. The center pushbutton in the out position shall indicate that the channel is monitored in the headset and a white light shall illuminate a translucent channel identification tab. The pushbutton in a depressed position shall transfer headphone monitor to speaker monitor and the switch light shall illuminate yellow. The brilliance shall be controlled by the subgroup's dimmer control.

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3.3.3.4.2.3 RSM transmitter select and receive switch. The top pushbutton when depressed shall indicate that the channel is selected by the operator for transmitter keying and receiver monitoring and a green light shall be on at full brilliance. The pushbutton in the out position shall indicate that the channel is not selected at the operator control subgroup and when an amber light is on at a nominal half-brilliance shall indicate that the channel transmitter is selected for keying at another operator control subgroup. When the channel transmitter is not selected by any operator, the green and amber lights shall be out and the channel transmitter shall be available for selection. When the channel transmitter is selected at another operator control subgroup and an attempt is made to select the channel transmitter, the amber light shall be on at full brilliance and a pulsating audio tone shall be in the headset or speaker at the operator control subgroup attempting the selection. Control circuitry shall be provided to mute the channel in the AN/FSA-32 speaker amplifier circuit when the channel transmitter is keyed.

3.3.3.4.3 Private interphone switch module (PISM). Each PISM shall contain three pushbutton switch lights for control of three independent interphone channels.

3.3.3.4.3.1 PISM call switch. The pushbutton depressed shall indicate that the channel is selected for transmission and a red light shall come on at full brilliance at the called position. The pushbutton out shall indicate that the channel is not selected and a white light shall be on to provide channel identification illumination. The depressed position shall transfer the interphone speaker monitor to headset operation. The red light on at full brilliance (3.3.3.3.10.1) with the pushbutton out shall indicate an incoming call and the calling control subgroup.

3.3.3.4.3.2 Converted PISM. The PISM shall have the facility to be converted into a position monitor switch (PMS) by simple mechanical adjustments and color light diffuser change. The adjustments shall be limited to disconnecting and connecting straps. Screw-type lock and connections are permissible. The PMS switches shall have the option for lock and momentary operation by installation or removal of a clip and pin. One pushbutton selector switch shall be used for each control subgroup's monitor channel.

3.3.3.4.3.3 PMS and PISM interchangeability. The PMS module shall be interchangeable with the interphone PISM in the interphone receptacles without damage to the switch, interphone, and position monitor circuits. Programming shall provide the monitor signal to the PMS. The operator control subgroup dimmer control shall control the identification illumination of the PMS switches. The PMS switch shall contain an amber light with removable print-on tabs and back lighting for channel identification.

3.3.3.4.4 Headset transfer switch. Each of the equipment's three types of jackboxes shall contain a switch for transferring one set of headset jacks to leased landline equipment and simultaneously transferring radiophone and interphone monitor to control subgroup speaker (see 3.3.3.9).

3.3.3.4.5 Control subgroup speaker mute switch. Each control subgroup speaker amplifier shall contain a lock-type pushbutton switch-light to reduce the speaker level by a pre-set amount (see 3.3.3.6) with an option connection for effecting this reduction automatically with transmitter key operation.

3.3.3.4.6 Foot switch. Each operator control subgroup with radiophone channels shall have a moveable foot switch for keying radiophone transmitters. The foot switch shall be type SA-585/FSA-17.

3.3.3.4.7 AN/FSA-32 overhead speaker(s) mute switch(es). Each control tower operator control subgroup shall have a Switching Panel SB-1281/FSA-32 which contains eight spring-loaded switches to reduce the speaker level individually by a pre-set amount and one switch to reduce the speaker level of all speakers simultaneously (see 3.3.3.10).

3.3.3.4.8 Incord microphone switch. Each headset shall have an incord lock-type switch for removing the microphone form the console amplifier (see 3.3.3.10).

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3.3.3.4.9 Handheld-microphone key switch. Each control tower operator control subgroup having a radiophone function shall be provided with a handheld microphone containing a switch for keying transmitter(s).

3.3.3.4.10 Simultaneous selection. Simultaneous selection and operation of radiophone and interphone channels and landlines in any combination within the limits of channels programmed shall be incorporated in the design. This operation shall be accomplished without degradation of the equipment.

3.3.3.4.11 Pushbutton switch illumination. Each pushbutton switch shall be illuminated and contain a removable switch cap with opaque sides and edges and with a removable clear plastic tab insert, a removable clear write-on plastic tab, and a thin white light diffuser with write-on capability.

3.3.3.4.12 Switch lamp color illumination. The lamp color illumination and signal indication for the pushbutton selector switches shall be as shown in TABLE I.

TABLE I. Switch lamp color illumination.

Switch module	Switch and position	Color	Function
PISM	Each off (2 lamps)	White	Identification illumination
	Each off (2 lamps)	Red	Incoming call
PMS	Each off (2 lamps)	White	Identification illumination
	Each depressed (2 lamps)	Yellow	Position selected for monitor
RSM	Top off	No light	Not selected by anyone
	(2 lamps)	Amber	Selected by other position
	Depressed	Bright green	Selected for transmitting
	Depressed	Bright amber	Overridden by supervisor control subgroup
	Center (2 lamps) off	White	Identification illumination
	Depressed	Yellow	Selected for speaker monitor
	Bottom off (2 lamps) off	No light	Not selected
	Depressed	Bright amber	Incoming signal
		Low amber	Selected for monitor
	Depressed	Bright amber	Incoming signal

All unprogrammed radiophone and interphone switches shall have no illumination regardless of switch position.

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3.3.3.5 Console amplifier unit (CAU). The CAU shall consist of two cases (shells), a console amplifier module, and a volume unit (VU) meter module. The CAU shall perform the same function as the console amplifier unit of the AN/FSA-58(XN-1)(V) except where otherwise specified herein.

3.3.3.5.1 Mounting and connections. The CAU cases shall be capable of mounting together vertically 177.8 mm (seven-inches) wide by 133.4 mm (5.25-inches) high and horizontally 133.4 mm (5.25-inches) wide and 177.8 mm (seven-inches) high with the identification markings upright in each situation. The cases shall contain guides and receptacles for accepting the console amplifier and VU meter modules and shall contain terminals for interface connections with the Landline Selector Group Console OA-7621(V)/FSA-52(V).

3.3.3.5.2 VU meter module. The VU meter module shall contain only the VU meter, the edge-lit panel, and illumination source.

3.3.3.5.3 Console amplifier module (CAM). The CAM shall contain mechanically keyed plug-in microphone, headphone and monitor amplifiers, and plug-in relays. The amplifiers shall be combined with a control panel into one plug-in assembly. The CAM shall provide connection to the landline selector control console of the OA-7621(V)/FSA-52(V), and shall include the interphone call relay. The operation or removal of any plug-in amplifiers shall not adversely affect the operation of the other circuits of the equipment. The busy tone generator shall be located in the CAM.

3.3.3.5.3.1 Microphone amplifier. The microphone amplifier shall have a 50-Ohm input transformer for accepting the H-170/U headset microphone and an output transformer of less than six Ohms balanced to ground. Compression- and expansion-type amplifier circuits are prohibited in the microphone amplifier. With the audio level control set at maximum and with an eight millivolt signal input, the microphone amplifier shall provide a zero dBm signal into a 600-Ohm line. There shall be less than 2 dB signal voltage variation with the load changed to 10 Ohms and without change in level control setting. The amplifier shall provide up to +6 dBm signal into a 600-Ohm load with a 0.8 millivolt input signal.

3.3.3.5.3.2 Headphone amplifier. The headphone amplifier shall have input and output transformers. The amplifier input shall be 50 Ohms balanced to ground and the output shall be 100 Ohms unbalanced. With an eight millivolt signal input and headphone level control set at maximum, the amplifier shall provide a minimum 0.8 volt signal into a 100 Ohm load.

3.3.3.5.3.3 Monitor amplifier. This amplifier shall be linear and shall have a transformer output of less than six Ohms balanced to ground. There shall be three inputs: a) one from the microphone amplifier output, b) one from the headphone amplifier output, and c) one from the speaker amplifier output. The input circuit design shall provide a minimum of 80 dB signal isolation between the three inputs. The monitor amplifier output signal shall be 0.8 vrm's into a 200 Ohm load when the microphone amplifier is providing a 0.8 vrm's output signal, when the headphone amplifier is providing a 0.03 vrm's output signal, and when the speaker amplifier is providing a 0.3 vrm's output signal. No input from the microphone amplifier shall be provided unless some communication mode has been selected at that position.

3.3.3.5.3.4 CAU control panel assembly. The control panel assemblies shall contain headphone and dimmer controls, a VU meter, transmitter key and interphone relays, edge-lighted panel to identify controls, receptacles for the plug-in amplifier cards, and wire terminals for connection to the OA-7621(V)/FSA-52(V) Landline Selector Control Console. The microphone amplifier level control shall be in accordance wiht 3.3.3.3.12. The VU meter shall have internal illumination varied by the control panel dimmer control. Light emission from the VU meter shall be restricted to reflections from the meter dial. The interphone call relay shall be located in these assemblies.

3.3.3.5.3.4.1 Subgroup dimmer control. This control shall vary the light intensity of all edge-lighted panels and switch identification illumination lamps in an operator control subgroup from off to full brilliance.

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3.3.3.6 Speaker amplifier unit. The speaker amplifier unit shall consist of a case and an amplifier module and shall perform the same functions as the speaker amplifier unit of the AN/FSA-58(XN-1)(V), except where otherwise specified herein.

3.3.3.6.1 Unit case. The unit case shall be fabricated in accordance with FIGURES 1 and 2 with a depth of no more than four inches overall with connectors unless otherwise approved by the procuring activity. The unit case shall contain all the guides, connectors, harnesses, receptacles, and locking devices for accepting the speaker amplifier module. The unit case shall provide 10 percent spare pins, connection to the landline selector control panel of the OA-7621(V)/FSA-52(V), and an optional connection to an external speaker. One 101.6 mm (four inch) speaker, Type LS-445/U in accordance with MIL-L-12606 shall be provided (for external use) with each speaker amplifier unit.

3.3.3.6.2 Speaker amplifier module. The speaker amplifier module shall contain the speaker, speaker amplifier circuitry, mute switch, mute level control, volume control, speaker audio transfer relay, edge-lighted panel, wiring for connection for automatic mute with transmitter keying, and mechanical locks. The interphone call relay shall be removed form the speaker amplifier module. A minimum of 10 percent spare contacts shall be provided. Circuitry shall be provided for optional connection of the external speaker and disconnection of the internal speaker.

3.3.3.6.2.1 Speaker amplifier. The speaker amplifier shall have an input of not more than 50 Ohms balanced and shall be modified to incorporate the level control in the input circuit. The amplifier shall provide 600 milliwatts into 3.2- and 8-Ohm loads with the amplifier level control set at mid-range and with input signals from any of the equipment circuits or channels at operating levels. An optional connection shall be provided for the automatically reduced speaker level (mute) operation when a radiophone channel is keyed by the operator.

3.3.3.6.2.2 Front control panel. This panel shall contain: a Quam Nichols Co., Type 25A070T (or equivalent), 3.2-Ohm speaker with 3.175 mm (0.125 inch) wire mesh cover, a mute switch light with yellow light, a reversible edge-lighted panel for horizontal or vertical unit mounting, mounting locks, and a volume control.

3.3.3.6.2.3 Printed circuit (PC) amplifier card. A keyed PC card shall contain a receptacle for a speaker to headset transfer relay, mute level control to pre-set the level of muting by the speaker mute switch light, and the speaker amplifier. The interphone call relay shall be removed and incorporated in the console amplifier module as specified in 3.3.3.5.3.

3.3.3.7 Radiophone switch unit (RSU). The RSU shall contain a unit case and five RSM's.

3.3.3.7.1 RSU unit case (shell). The unit case shall be fabricated in accordance with FIGURES 1 and 2 and the depth shall be no more than 101.6 mm (four inches) overall with connectors unless otherwise approved by the procuring activity. The unit case shall contain all the switch guides, connectors, receptacles, lock fastener receptacles to accept five RSM's, and 10 percent spare pins in the connector receptacles.

3.3.3.7.2 RSM. The operation of each RSM shall be in accordance with 3.3.3.4.2. Circuitry shall be provided to mute the speaker-monitored channels selected for transmission during keying. The channels selected for speaker monitor and not keyed shall not be affected except for the reduced level mute switch. Also, circuitry shall be provided for muting the channels keyed in the overhead speaker connected to the AN/FSA-32. The true sidetone shall be reduced in the headphone 12 dB from the level of other received signals.

3.3.3.8 Private interphone (monitor) switch unit (PISM). The PISM shall contain a unit case and five PISM's.

3.3.3.8.1 Unit case. The unit case shall be fabricated in accordance with FIGURES 1 and 2 and the depth shall be no more than 101.6 mm (four inches) overall with connectors unless otherwise approved by the procuring activity. The unit case shall contain all the switch guides, connectors, receptacles, and lock fasteners to accept five PISM's or PMS's and 10 percent spare pins in both plug and receptacle.

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3.3.3.8.2 PISM. Operation of each PISM shall be as specified in 3.3.3.4.3. The PISM shall contain circuitry to provide:

a. Reduced sidetone level to headphone amplifier
b. Channel mute in the speaker unit
c. Identification illumination
d. Option for speaker or headphone monitor on call-up
e. Modification of interphone call circuit to include solid state switch and strap-type connection for optional automatic instantaneous interphone override response to an interphone call
f. Automatic transfer to headset of channels when the call is answered

3.3.3.9 Jackboxes. There shall be three types of jackboxes with each system, as specified in 3.3.3.9.1 through 3.3.3.9.3.

3.3.3.9.1 Jackbox J-( )/FSA-17. This jackbox shall contain two headset jacks, one headset transfer lever switch, one transfer indicator blue light, and a case with connectors and a facility for mounting in any desired location in the area of operator position. The jackbox shall provide the following features:

a. Two sets of parallel headset jacks
b. Transfer one set of headset jacks to leased landline equipment by activating the two-position lever switch
c. Automatic transfer of radio and interphone monitoring to speaker monitoring when headset jacks are transferred to leased landline equipment and headset is removed from the RADIO ONLY set of jacks
d. Automatic illumination of a blue light when the headset jacks are transferred to leased landline equipment
e. Continuous operation of radiophone and interphone in the RADIO ONLY headset jacks when the headset is plugged-in
f. Automatic transfer of radiophone and interphone to speaker when the headsets are removed from both sets of jacks

Jackbox J-( )/FSA-17 shall provide switching of the jacks marked RADIO LANDLINE to OA-7621(V)/FSA-52(V), Landline Control and to permit the jacks marked RADIO ONLY to remain in radio and private interphone communications for dual operation from the same jackbox. The switch position shall be labeled SEPARATE and COMMON in the jackbox.

3.3.3.9.2 Modular unit jackbox. This jackbox shall perform the same functions as in 3.3.3.9.1 but shall be mounted in a unit shell fabricated in accordance with FIGURES 1 and 2 and less than 101.6 mm (four-inch) depth unless otherwise approved by the procuring activity. The jackbox insert and circuitry for the edge-lighted panel shall be provided. The edge-lighted panel shall provide for vertical or horizontal mounting in the vertical plane.

3.3.3.9.3 Control tower operator jackbox. This jackbox shall contain the same features as the modular unit jackbox described in 3.3.3.9.2 and a third set of jacks for a hand-held microphone with a transmitter key switch to provide the following features:

a. The transmitter key footswitch shall be in parallel for keying with the hand-held microphone switch
b. The hand-held microphone plugs and jacks shall be of a fast connect and disconnect similar to the headset type
c. The jacks for the hand-held microphone shall be mechanically different than those of the headsets to prevent inadvertent plug-in to headset jacks and vice versa
d. The hand-held microphone shall be limited to radiophone and interphone operation with side tone muted in the subgroup speaker. Other radiophone channels and interphone channels shall be continuously monitored in the operator subgroup speaker

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3.3.3.10 Headset. Two headsets H-170/U shall be provided with each operator control subgroup. The headset shall be a noise-canceling microphone, Roanwell Corporation model RN-1C, or equivalent.

3.3.3.11 Hand-held microphone. The hand-held microphone shall be provided as required in 3.3.3.9.3 and as follows:

a. The noise canceling feature shall provide a minimum of 14 dB noise cancelation
b. The hand-held microphone audio shall be connected to the microphone amplifier without switching
c. A hand-held microphone with spring return key switch shall be incorporated in the control tower operator control subgroup

3.3.3.12 Footswitch. The foot bar switch shall be the two-position Foot Switch SA-585/FSA-17. The up position shall be off. The spring-return or depressed position shall be radio transmitter key for these transmitters selected, and shall cause the keyed radiophone channels to be muted in the overhead speakers and true sidetone to be received in the headphones.

3.3.3.13 Connector harnesses. Connector harnesses shall be provided as required for the equipment specified in the contract or order. Connector cable harness wire insulation shall be 1000 volts. A bracket for mounting each cable terminal block connector harness in the rear of standard 19-inch and 24-inch equipment cabinet spaces shall be provided. The brackets shall provide for a minimum of 25.4 mm (one-inch) spacing between terminal blocks. FIGURE 3 shows standard 19-inch Relay Rack and panel dimensions.

3.3.3.13.1 Cable terminal block connector harness. This harness shall be similar to J-2865/FSA-52(V) and shall provide the necessary connections to the CAU, two RSU's, one PISMU and one speaker unit. It shall also provide connector receptacles to allow parallel connections form two additional RSU's and one additional PISMU. This harness shall provide for wirewrap termination of cables from the control central subgoup. Flexible conductors shall be used in the connector harness with solder connection at the terminal block. The connector harness shall be five-feet long. Holders for removable print-on tabs shall be provided to identify control position, cable functions, and connector receptacles in addition to markings required for terminal block terminals in accordance with MIL-E-16400.

3.3.3.13.2 Add-on cable terminal block connector harness. This harness shall contain necessary terminations and connections to allow for two additional RSM's or one PISMU. It shall contain a connector receptacle for an additional add-on harness.

3.3.3.14 Landline dial console units. A dial console unit shall be provided with dimensions of 133.4 mm (5.25 inches) by 88.9 mm (3.50 inches) and by no more than 101.6 mm (four inches) deep overall including connectors. The packaged dial console unit shall perform the exact same functions as and shall be electrically interchangeable with the existing unit 58 of the OA-7621(V)/FSA-52(V). Fabrication detail and schematic diagrams for the existing dial console units are provided in FIGURE 4 (six sheets).

3.3.3.15 Landline selector and control console unit. A landline selector and control console unit shall be provided with dimensions 133.4 mm (5.25 inches) by 88.9 mm (3.50 inches) and by no more than 101.6 mm (four inches) deep overall including connectors. The landline selector and control console unit shall perform the exact same functions as and shall be electrically interchangeable with the present unit 59 of the OA-7621(V)/FSA-52(V). The landline selector and control and dial console connector harness and terminal block assembly J-2865/FSA-52(V) shall be modified to provide the required cable harness length and connectors. One modified connector harness and terminal block assembly shall be provided with each packaged landline selector and control console unit. FIGURE 5 (six sheets) contains fabrication detail and schematic diagrams for the existing landline selector and control console unit.

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3.3.3.16 Console unit mounting bracket kits. There shall be four types of brackets for mounting the console units in standard equipment rack panel spaces of 19 and 24 inches as specified in 3.3.3.16.1 through 3.3.3.16.5.

3.3.3.16.1 Mounting bracket kit type A. This kit shall mount three units with the 133.4 mm (5.25 inch) dimension horizontally in a 19-inch panel space.

3.3.3.16.2 Mounting bracket kit type B. This kit shall mount five units with the 88.9 mm (3.50 inch) dimension horizontally in a 19-inch panel space.

3.3.3.16.3 Mounting bracket kit type C. This kit shall mount four units with the 139.7 mm (5.50 inch) dimension horizontally in a 24-inch panel space.

3.3.3.16.4 Mounting bracket kit type D. This kit shall mount six units with the 88.9 mm (3.50 inch) dimension horizontally in a 24-inch panel space.

3.3.3.16.5 Mounting bracket kits. Unless otherwise specified in the contract or order, mounting bracket kits shall be provided for each control subgroup as follows:

Type	Op. Con. Subgroup Qty.	Sup. Con. Subgroup Qty.
A	3	4
B	1	2
C	2	3
D	1	2

3.3.3.17 Remote operator control subgroup operation. This operator control subgroup shall operate using the cable size specified herein without loss of performance when located up to 6000 feet from control central and up to five miles using a local power supply. Proper operation of the remote operator control subgroup shall be accomplished without splitting pairs of conductors in the cables. Installation cable conductor size will be No. 22 gauge solid for distances of up to 1000 feet and No. 19 gauge for distances greater than 1000 feet.

3.3.4 Supervisor control subgroup. This control subgroup shall contain all the circuitry, switches, and components that comprise the supervisor control subgroup for control of all radiophone, interphone, monitor, and landline channels available to the equipment, and that which is specified in 3.3.4.1 through 3.3.4.5.

3.3.4.1 Status switch unit (SSU). The SSU shall consist of a unit case and five Status Switch Modules (SSM) for channel programming and control subgroup display. The SSU design shall provide for the replacement of one SSM by a Supervisor Special Control Module (SSCM) (see 3.3.4.1.3).

3.3.4.1.1 Unit case. The unit case shall be fabricated in accordance with FIGURES 1 and 2 except the depth shall be no more than 101.6 mm (four inches) overall with connectors unless otherwise approved by the procuring activity. The unit case shall contain all the switch guides, connectors, receptacles, and lock fastener receptacles to accept five SSM's with a minimum of 10-percent spare pins in the connector plugs and receptacles.

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3.3.4.1.2 SSM. The SSM shall be fabricated in accordance with 3.3.3.4.1 except that the push-button switches shall be spring return. The SSM shall provide for displaying channels programmed to an operator control subgroup and indicating the operator control subgroup that has a channel selected for transmitting. When the SSM is used for channel program display, each of the three pushbutton switches shall indicate a channel. When the SSM is used for operator control subgroup display, each of the three pushbutton switches shall indicate an operator control subgroup. When a pushbutton switch is depressed in the channel program display, a light shall turn on at full brilliance in the operator control subgroup display, indicating the operator control subgroup that has selected the channel for transmitting. When a pushbutton switch is depressed in the operator control subgroup display, lights shall turn on at full brilliance at the channel program display, indicating all the channels programmed to the operator control subgroup that is associated with the depressed switch. There shall be no damage to the equipment components in the event that one or more display bushbutton switches are depressed. The interconnection in the wiring on the terminal block assemblies at the supervisor control subgroup shall determine the type display of the SSU. There shall be no degradation of equipment operation with the removal and operation of the SSM(s).

3.3.4.1.3 SSCM. The SSCM shall be fabricated in accordance with 3.3.3.4.1. The top switch shall be for channel in-use or selected display control. In the depressed position, channels selected for transmission shall be identified in the channel program display modules. The center switch shall be the supervisor all interphone call switch. In the depressed position, the supervisor control subgroup shall initiate a call to all operator control subgroups that are programmed interphone to the supervisor subgroup. At each called operator control subgroup, the interphone call operation shall be the same as a call initiated on an individual call basis. The bottom switch shall provide in the supervisor control subgroup, a red light and audio alarm indication of a blown fuse or fault in the equipment. In the depressed position, the audio alarm shall be silenced at the supervisor control subgroup and a white light shall indicate alarm cutoff. The interconnections in the wiring on the terminal block assemblies at the supervisor control subgroup shall determine the receptacle in the (SSU) for accepting the SSCM. The SSCM and SSM modules shall be interchangeable in the SSU without damage to the module or the display circuits and without degradation of equipment operation. There shall be no degradation of equipment operation with the removal or operation of the SSCM.

3.3.4.2 Supervisor RSM operation. The RSM operation in the supervisor control subgroup shall be the same as at the operator control subgroup except that when the radiophone transmit-select switch is operated, the supervisor control subgroup will seize control. The pre-empted operator control subgroup and any other operator control subgroup attempting control shall receive the busy indications. Radiophone receiver monitor capability shall be retained at the operator control subgroups.

3.3.4.3 Solid state switching. The supervisors control subgroup shall incorporate solid state switching for radiophone transmitter selection and control. All interphone call switching shall be solid state.

3.3.4.4 Connector harnesses. Connector harnesses shall be provided for this subgroup as specified in 3.3.3.13.

3.3.4.4.1 Cable terminal block connector harness. This harness shall be provided for this subgroup as specified in 3.3.3.13.1.

3.3.4.4.2 Add-on cable terminal block connector harness. This harness shall be provided for this subgroup as specified in 3.3.3.13.2.

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3.3.4.5 Switch light color illumination. The light color illumination for the pushbutton selector switches shall be as shown in TABLE II.

TABLE II. Switch light color illumination.

Switch module	Switch & Position	Color	Function
SSM (channel program	Each off (2 lamps)	White	Identification illumination
	Each off (2 lamps)	Amber	Channel programming or position controlling channel transmitter
SSCM	Top off (2 lamps)	Dim white	Identification illumination
	Depressed	Bright white	Identification illumination
	Center off (2 lamps)	Dim white	Identification illumination
	Depressed	Bright white	Identification illumination
	Bottom off (2 lamps) off	No light Red	No system fault System fault alarm
	Depressed	Bright white	Fault alarm cut-off

3.3.5 Control central subgroup. The control central subgroup shall contain all the circuitry, switches, and components that comprise the control central subgroup for control of all radiophone, interphone, monitor, and landline channels available to the equipment, and that specified in 3.3.5.1 through 3.3.5.11.

3.3.5.1 Transmitter-operator select/control. The transmitter-operator select/control circuits of the AN/FSA-58(V) shall be packaged using solid state switching circuits. The circuits shall provide for 60 operator control subgroups and radiophone channels. Except where otherwise specified herein, the same operational features as in the AN/FSA-58(V) (XN-1) shall be provided. Positive lockout to preclude more than one operator control subgroup from selecting and modulating a transmitter simultaneously shall be retained. Each transmitter control unit shall provide for control and termination of 10 transmitters. The unit shall be 88.9 mm (3.50 inches) high, 254 mm (10-inches) deep (overall with terminal blocks) and shall mount in a 19-inch wide standard equipment rack. A plug-in transformer with retainers and 600-Ohm output balanced to ground and optional simplex keying shall be provided. A plug-in relay with retainer and contacts for main and auxiliary key and for audio switching shall be provided for each transmitter channel.

3.3.5.1.1 Transmitter/select circuits. The following features shall be provided:

a. Up to 60 operator control subgroups accessing a radiophone channel transmitter on a lock-out basis, upon selection, to all other operator control subgroups in an equipment installation
b. A positive lockout to prevent simultaneous selection and modulation by more than one control subgroup's operator
c. A lockout to provide for a busy signal to any other operator control subgroup attempting to select the channel

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d. Supervisor override with busy signal visual and audible signals to the overridden operator control subgroup
e. In-use indication to all other operator and supervisor control subgroups upon selection of a transmitter
f. Two-wire control for both transmitter key and auxiliary key
g. Supervisor indicate transmitter control function
h. Option for select to lockout and key to lockout on an individual channel basis
i. Lockout option accomplished with installation requiring wirewrap or screw type connection(s)

3.3.5.2 Receiver amplifier indicator unit. The receiver amplifier indicator unit of the AN/FSA-58(V) shall be packaged to incorporate the line amplifier functions of the Amplifier Group, OA-4391/FSA-17, for main and standby receiver channels with control from the Transfer Control Group, AN/FSA-47(V). The unit shall provide for 10 main and 10 standby channels and thumbwheel switch selection for monitor. The unit shall be 88.9 mm (3.50 inches) high, 254 mm (10-inches) deep (overall with terminal blocks), and mount in a 19-inch standard equipment rack panel space. One input and one output terminal block shall be provided. One receptacle with plug-in connector shall be provided for connection to the monitor panel.

3.3.5.2.1 Receiver amplifier indicator module. The receiver amplifier indicator module of the AN/FSA-58(V) shall be packaged to incorporate the line level compensate function of the OA-4391/FSA-17 and to provide a constant signal voltage of 0.8 volts with one to 60 operator control subgroups connected with no more than 2-dB change in output level with a minus 20-dBm input signal. The amplifier shall be able to compensate automatically for the varying load at the output without requiring adjustment. The module shall also be able to compensate by means of a screwdriver adjustment at the front panel for a variable input level of minus 20 dBm to plus 10 dBm for both main and standby channels. A voice-operated circuit in the amplifier shall illuminate amber lights (one on the module, one at each programmed operator control subgroup, and one at the supervisor control subgroup) whenever there is an incoming signal on the channel above a pre-set level. The amber light shall appear on each amplifier module. The sensitivity of the voice-operated transmitter keyer (VOX) indicate circuit shall have a variable threshold control to permit an input signal level of minus 25 dBm to illuminate the lights. There shall be no degradation of the voice signal circuit by operation of the VOX indicate circuit.

3.3.5.3 Radiophone program unit. This unit shall enable the control central subgroup to obtain the rapid and flexible programming of control of receivers and transmitters and shall indicate status for each of the equipment's radiophone channels in operator control subgroups and the supervisor subgroup. Each unit shall provide programming to 10 RSM's in each of five operator control subgroups from a selection of 20 radiophone channels and shall provide the supervisor status of these channels and operator control subgroups. Connectors, Type 22 male pin contacts Methode P/N GPXMD-622M, or equivalent shall be incorporated in the program panel. Wire wrap terminal blocks shall be provided for terminating incoming operator and channel cables, status functions to supervisor, and for parallel connection of radiophone program units for selection from additional channels and more channels to an operator control subgroup. Individual terminal blocks shall be provided for each operator control subgroup and supervisor status function. The unit shall contain a hinged dust cover front panel. Tab holders for removable print-on tabs for identifying control position and channels shall be provided on the front panel. The radiophone channels shall be numbered. The radiophone program unit shall be 133.4 mm (5.25 inches) high, 254 mm (10-inches) deep (overall with terminal blocks), and mount in a 19-inch standard equipment rack. Tab holders shall be provided for removable print-on tabs to identify the operator control subgroup, switch number, and channel for each terminal block. Pins on the terminal blocks shall be identified. The program panel shall be recessed to permit two or more program units to be mounted adjacent to each other and enable patching between panels.

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3.3.5.3.1 Radiophone program patch cords. The color of radiophone patch cords shall be green and the plugs shall be keyed to allow only the proper connection. Patch cords and lengths shall be provided with each radiophone program unit as follows:

Qty	Lengths to:
60	Span the extremes of a single radiophone program unit (20 channels, 10 channels/operator)
35	Span the extremes of two radiophone program units in parallel (40 channels, 20 channels/operator)
15	Span the extremes of three radiophone program units in parallel (60 channels, 30 channels/operator)

3.3.5.4 Private interphone program unit. Private interphone programming shall be provided for rapid and flexible programming of the direct private interphone lines and operator control subgroup monitor channels. Connector, Type 22 male pin contacts Methode P/N GPXMD-622M, or equivalent shall be incorporated in the program panel. Each unit shall contain eight segments, each with 15 channel receptacles. Tab holders for removable print-on tabs to identify operator control subgroup, private interphone channel subgroup and monitor channel subgroup shall be provided for each segment. The unit shall be 133.4 mm (5.25 inches) high, 254 mm (10-inches) deep, and mount in a 19-inch standard equipment rack. The program panels shall be recessed. Each segment shall be separated and identified for rapid and efficient programming. Each program unit shall contain a hinged dust cover with quarter-turn fasteners and shall open from left to right. The recessed panel shall permit two or more program units to be mounted adjacent to each other and enable patching between panels. The interphone program unit shall accept patch cords from the position monitor program units. Each interphone program unit segment shall have an individual terminal block.

3.3.5.4.1 Interphone program patch cords. These patch cords shall be red and shall be keyed to allow only proper connection. The number and lengths of patch cords furnished with each private interphone program unit shall be as follows:

Qty	Lengths to:
54	Span the extremes of four adjacent segments
24	Span the extremes from six adjacent segments to four adjacent segments of a second panel separated by a monitor program unit
12	Span the extremes of two interphone program units mounted together with a monitor program unit between the two

3.3.5.5 Monitor program unit. This unit shall provide the subgroup recording audio for programming to the interphone program units for monitoring the operator control subgroups. Each subgroup monitor program unit shall contain six groups of receptacles. Each group shall contain twelve parallel receptacles. The program panel shall be 44.5 mm (1.75 inches) high, 254 mm (10-inches) deep, and mount in a standard 19-inch equipment rack. The program panel shall be recessed for flush mounting with the interphone program unit. There shall be wirewrap terminal posts for incoming cables. A hinged dust cover shall be provided with a quarter-turn fastener and shall open from left to right. Connector, Type 22 male pin contacts, Methode P/N GPXMD-622M, or equivalent, shall be incorporated in the subgroup monitor program panel.

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3.3.5.5.1 Monitor program patch cords. These patch cords shall be yellow and shall be keyed to allow only proper connection. One end shall have the same number of jacks as the interphone patch cord. The number and lengths of patch cords furnished with each subgroup monitor program unit shall be as follows:

Qty	Lengths to:
48	Span the extremes from two adjacent segments of monitor program unit and nearest four segments of an adjacent interphone program unit
18	Span extremes from the segments of a monitor program unit to the extremes of an adjacent interphone program unit

3.3.5.6 Monitor panel unit. The monitor panel unit of the AN/FSA-58(V) shall be packaged on a 5.25 x 19-inch panel and shall provide circuitry and components as follows: An audio signal generator, a VU meter for equipment circuit monitor, a pulsating visual and audible fault alarm indication at control central subgroup and at the supervisor control subgroup, fault alarm activation from ancillary equipments external to the AN/FSA-58(V), a front panel selector switch for selecting any one of ten receiver amplifier indicator units for VU meter monitor, switching to provide speaker or headphone monitor of the VU meter signal without change in the VU meter signal level, a speaker amplifier with 0.5-watt output and a plus 6 dBm output for headphone with a minus 20 dBm input and a frequency response of 300 Hz to 3000 Hz ±1 dB, a 101.6 mm (four-inch) speaker in accordance with MIL-L-12606, a front panel speaker volume control, a front panel headphone volume control, a technician-type variable control with lock to set the audio alarm level and to bypass the front panel speaker volume control, an audio alarm level and to bypass the front panel speaker volume control, and audio fault alarm cutoff switch, a red lens (Type LH75C14RD in accordance with MIL-L-3661) lampholder with flashing light indicating a fault in the AN/FSA-58(V) system and ancillary equipment connected for automatic fault monitor, circuitry for automatic reset when a fault is corrected, headphone jacks to accept the receiver circuit of the H-170/U headset, and a spring-return switch for the fault alarm signal generator output level adjust. The audio signal generator shall provide a 600-Hz tone with 0.5-second pulses spaced 0.5-seconds apart when a fault appears in the system. There shall be no degradation of equipment performance by the operation or removal of the monitor panel. The audio signal generator, the fault alarm detector, and the speaker amplifier shall be on plug-in cards.

3.3.5.7 Power supply unit. This unit shall consist of a rectifier/charger, battery, main distribution fuse block panel, batteries and fuse/power distribution panel(s). This unit shall provide: plus 25 VDC (nominal) for the system full operation power requirement, battery charge, battery eliminator filtering, charge fail alarm relay, equalizer charge, equalizer charge timer, and over voltage cutout.

3.3.5.7.1 Rectifier/charger. A rectifier/charger shall be provided for full system operation current load requirements as follows:

Full load	Rectifier charger
Up to 9.6 amperes	Lorain Model M12F25
Between 9.6 to 20.0 amperes	Lorain Model M25F25
Between 20.0 to 25.6 amperes	Lorain Model M32F25
Between 25.6 to 40.0 amperes	Lorain Model M50F25
Between 40.0 to 60.0 amperes	Lorain Model M75F25
Between 60.0 to 80.0 amperes	Lorain Model M100F25

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Each rectifier charger shall be provided with Lorain accessories as follows: charge fail alarm relay, battery eliminator filtering, and equalizer charge timer. There shall be no degradation of system full operation with battery connected or disconnected.

3.3.5.7.2 Main distribution fuse block assembly. This assembly shall provide for fuse block disconnects for the rectifier/charger battery (both sides) and three power outputs as follows:

AM/FSA-58 systems full load circuit	Distribution fuse panel
0-25 amperes	Lorain, No. 1124AS, Modified, with six 0-30 ampere fuse blocks and fuses
26-55 amperes	Lorain, No. 1125AS, Modified, with three 0-30 ampere fuse blocks and three 31 to 60 ampere fuse blocks and fuses

3.3.5.7.3 Fuse/power distribution panel. The front panel shall provide for fused power distribution with open fuse indication to 40 individual circuits as follows: One fuse for illumination and one fuse for electronics and signalling circuits for each of five operator control subgroups, one fuse for each radiophone channel transmitter keying and receiver indicator amplifier for each of 20 radiophone channels, two for illumination of and two for electronics and signalling for the supervisor control subgroup, one for the monitor panel, and five spare fused power outputs. The panel shall contain receptacles for plug-in open fuse detector cards and fuse alarm plug-in boards. The front panel shall be hinged with captive knurled and slotted headscrew fasteners. The front panel shall contain: 40 indicating fuse holders in accordance with MIL-F-19207, Type FHL-18G-2; one green lens lamp holder in accordance with MIL-L-3661, Type 81410, with incandescent lamp, 28 Volts, 0.17-ampere, Type T-3-1/4 for POWER ON indicator; one red lens (Type LH75C14RD), lamp holder in accordance with MIL-L-3661, Type 81410 with incandescent lamp, 28 Volts, 0.17-ampere Type T-3-1/4 for panel fault ALARM indicator; one rotary step switch for open fuse indicator, with four indicating fuse holders in one row selected by each step for indicating fuse holder lighting and one OFF position; ON-OFF switch and holders for removable print-on tabs for identifying power use, the control subgroup, and the radiophone channel associated with each indicating fuse holder. The switches and switch positions and lamps shall be identified. The operation of the rotary switch and removal of lamps shall not cause any degradation of equipment operation. Plug-in open fuse detector cards and open fuse alarm cards shall be provided with each panel. Wirewrap post terminal blocks and a separate power input terminal block shall be provided. Holders for removable print-on tabs shall be provided to identify terminal blocks and terminal function. The panel shall be 133.4 mm (5.25 inches) high and mount in standard 19-inch wide rack panel spaces.

3.3.5.7.4 Battery operation. A battery shall be provided to furnish one hour of full system operation from full charge with rectifier/charger disconnected. The batteries shall be Gould National Battery, Type 3-DS5-50, or equivalent.

3.3.5.8 Amplifier-mixer-muter group AN/FSA-32. An AN/FSA-32 consisting of the following shall be provided, including speakers (not normally supplied with the AN/FSA-32), with each equipment configuration procured unless otherwise specified in the contract or order:

a. Two each, Power Supply PP2947/FSA-32 (see 3.3.1.4)
   (one operating, one spare)
b. Nine each, Amplifier, Audio Frequency AM-2983/FSA-32 (see 3.3.1.4)
   (eight operating, one spare)
c. Four each, Amplifier, Mixer AM-2984/FSA-32 (see 3.3.1.4)
   (three operating, one spare)
d. Four each, Panel, Switching SB-1281/FSA-32 (see 3.3.1.2)
   (three operating, one spare)
e. One each, Test Set Electronic TS-1664/FSA-32 (see 3.3.1.5)
f. One each, Cabinet, Electrical Equipment CY-3129/FSA-32 (see 3.3.1.4)
g. Ten each, Speakers LS-445/U (see 3.3.1.2)
   (eight operating, two spares)

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The amplifier-mixer-muter group AN/FSA-32 shall provide for control of overhead speaker monitoring of the control tower operator subgroup, of up to 20 audio channels with individual channel muting. The basic power requirements shall be 105 to 125 Volts, 50 to 60 Hz, single-phase. The SB-1281/FSA-32 and LS-445/U are part of the control tower operator control subgroup. The TS-1664/FSA-32 is part of the support and diagnostic subgroup. The unit shall have the following characteristics:

a. Number of channels: 20 inputs to eight outputs
b. Input impedance: each channel 1200 Ohms balanced
c. Output impedance: to each speaker 600 Ohms balanced
d. Power output: to each speaker two watts
e. Input level: each channel 0.8 Volts rms balanced
f. Frequency response: 300 to 3000 Hz ±3 dB
g. Distortion: less than five percent at rated output
h. Muting attenuation: 40 dB (min)
i. HI-LO attenuation: 15 dB
j. Interchannel isolation: 80 dB
k. Power requirements: 105 to 125 volts, 50 to 60 Hz, single-phase, 170 watts
l. Ambient temperature: 0° to 50°C

3.3.5.8.1 Operator switching panel SB-1281/FSA-32. This unit shall contain eight spring-loaded two-position lever-operated switches, that are connected to the HI-LO function of the speaker amplifier described in 3.3.5.8.3. When the switch is depressed, the HI-LO control line for the associated amplifier shall be grounded.

3.3.5.8.1.1 ALL Switch. An ALL switch shall be provided (actually two switches tied together mechanically). The ALL switch shall function in the same manner as the eight individual switches except that it shall place a ground on all the HI-LO central lines at the same time. The activation of any of the HI-LO switches, when depressed, shall reduce the output of any particular speaker and releasing the switch shall automatically return speaker output to its original level. Pressing the ALL lever switch down shall temporarily attenuate the output of all eight speakers. If an individual speaker switch is pressed down while holding down the ALL switch, that individual speaker shall be restored to full output while the others shall still be attenuated until the switches are released.

3.3.5.8.1.2 DIMMER control. A DIMMER control shall be provided to control the edge-lighted marking. These lighted markers shall indicate: OVERHEAD SPEAKERS: HI↑; LO↓; 1; 2; 3; 4; 5; 6; 7; 8; ALL; DIMMER. The lighted numbers, ALL, and DIMMER shall be above the control. In addition, a surface shall be provided above the 1 through 8 speaker switches on which pencil marking can be made and removed for marking of frequency.

3.3.5.8.1.3 Panel switching SB-1281/FSA-32. This switching panel and case shall be packaged for mounting in a standard 24-inch rack panel spacing. The height shall be 88.2 mm (3.47 inches).

3.3.5.8.2 Amplifiers.

3.3.5.8.2.1 Amplifier audio frequency AM-2983/FSA-32. The audio frequency amplifier AM-2983/FSA-32 shall provide for amplification of the audio signals and consists of four sections; input circuits, preamplifier, driver-output stage, and collector supply ripple filter. It shall provide two watts of audio output (into a 600-Ohms speaker) with an input level of 0.775 volts. The output of the amplifier shall be controlled by a potentiometer mounted on the front panel and marked LEVEL. The output of the amplifier shall be capable of being checked through use of a monitoring jack mounted on the front panel and marked OUTPUT MONITOR. Amplification of the unit shall also be controlled through the HI-LO switch located on panel, switching SB-1281/FSA-32. Activation of the HI-LO switch to LO shall reduce the gain of the amplifier AM-2983/FSA-32 by about 15 dB. Provision shall be made to reduce gain of the unit by grounding of the muting control line, which shall attenuate the input signal by approximately 40 dB. The power input to the unit shall be protected by a 1.5-ampere fuse that shall be accessible from the front panel. Speaker amplifier AM-2983/FSA-32 shall have overall dimensions of: 133.4 mm (5.25-inch) height, 76.2 mm (three-inch) width, 11.50-inch depth.

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3.3.5.8.2.2 Amplifier mixer AM-2984/FSA-32. The amplifier mixer AM-2984/FSA-32 shall provide for the mixing of five identical inputs into a common amplifier. As the input section attenuates the signal about 34 dB, the amplifier shall restore the signal to its original level. A balanced input shall be provided plus provisions for muting signal. In the presence of the muting signal, attenuation of approximately 40 dB shall be provided. The output shall be balanced to drive audio frequency amplifier AM-2983/FSA-32. One side of the balanced output shall be made available for monitoring from front panel test points, TP1, which is ground and marked 1 and TP2 which is audio output marked TEST. The unit overall dimensions are: 31.8 mm (1.25-inch) height, 130.2 mm (5.125-inch) width, 292.1 mm (11.50-inch) depth, 1212.86 cu cms (74 cubic inches) volume, and 0.4536 kg (one pound) weight.

3.3.5.8.2.3 Power supply PP-2947/FSA-32. Power supply PP-9847/FSA-32 upon the application of an input of nominal 115 VAC single phase, shall provide plus five Volts DC and minus 12 Volts DC to amplifier audio frequency AM-2983/FSA-32 and amplifier mixer AM-2984/FSA-32. The unit shall be protected by a three-ampere indicating-type fuse which shall be accessible fron the front panel. The input power shall be controlled by an ON-OFF switch marked POWER ON - OFF. On the front panel shall be a pilot lamp with green lens to indicate power on. Also on the front panel shall be a fuse holder for a one-ampere fuse to protect the minus 12-Volt output used to light the panel light. Provisions shall be made for independent adjustment of the plus five and the minus 10-Volt outputs of the unit. A relay shall be connected across the minus 12 Volts, which upon the loss of the minus 12 Volts, shall close a set of contacts. The relay may be connected to an alarm system (which shall not be a part of the AN/FSA-32).

3.3.5.8.2.4 Cabinet CY-3129/FSA-32. Cabinet CY-3129/FSA-32 overall dimensions shall be: 10.50-inch height, 19-inch width, and 16-inch depth. It shall contain, support, and protect the units that make up the amplifier-mixer-muter group AN/FSA-32 as described in 3.3.5.8 and provide same internal connection between units and for inputs and outputs to external units. Provisions shall be made for the circulation of air for cooling of the units enclosed by convection means.

3.3.5.8.2.4.1 Channel muting modification. A modification shall be provided in this cabinet to effect channel muting of selected channels in any combination during key operation.

3.3.5.8.3 Speaker. The loudspeaker supplied shall be LS 445/U in accordance with MIL-L-12606. A quantity of 10 each (eight operating and two spare) shall be provided with matching transformers.

3.3.5.8.4 Interchangeability. All plug-in units and PC cards shall be electrically and mechanically interchangeable and shall perform the same functions as existing AN/FSA-32 plug-in units and PC cards.

3.3.5.9 AN/FSA-47(V) transfer control group. The AN/FSA-47(V) transfer control group is a modular transfer control means that provides fast transfer of control functions between two similar systems or groups of circuitry, or conversely, can be used to quickly transfer the control of one common equipment between two remote and separated control locations. The transfer function shall be accomplished by depressing a button. This shall enable an operator to control the transfer function without having to call on any other person for assistance (particularly applicable in air traffic control operations). A controller may be issuing instructions to an aircraft when his transmitter breaks down; therefore, the operator will be able to immediately call up a hot standby transmitter pre-tuned to the same frequency by depressing a button on the AN/FSA-47(V) remote control panel. The total down-time in this case will be the length of time it takes the operator to realize that his transmitter is not functioning properly plus the time required to push the button. The system modules are associated in pairs: one pair consisting of a remote unit usually located in an operational area and a local unit usually located in a maintenance or equipment area. The AN/FSA-47(V) transfer control group consists of remote switching modules, local switching modules, a dimmer module, and a common power supply module. One AN/FSA-47(V) consisting of the following shall be provided with each equipment configuration procured unless otherwise specified in the contract or order:

a. One each, Cabinet, Electrical Equipment CY-4040/FSA-47 (see 3.3.1.4)
b. One each, Cabinet, Electrical Equipment CY-4041/FSA-47 (see 3.3.5.9.3.4)
c. Two each, Cabinet, Electrical Equipment CY-4042/FSA-47 (see 3.3.1.3)
d. One each, Panel, Dimmer C-4792/FSA-47 (see 3.3.1.3)

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e. Three each, Panel, Switching SB-1995/FSA-47 (see 3.3.1.4)
f. Three each, Panel, Remote Switching SB-1996/FSA-47 (see 3.3.1.3)
g. One each, Power Supply PP-3824/FSA-47 (see 3.3.1.4)

Items c, d, and f are part of the supervisor control subgroup.

3.3.5.9.1 Characteristics. The unit shall have the following characteristics:

a. Channel capacity: Nominally, 15 radiophone channels using transmitters and receivers in a one-for-one back-up ratio
   Minimum, five channels (one local and remote switching panel, one power supply, one dimmer panel)
   Maximum, 25 channels (five local and five remote switching panels, one power supply, one dimmer panel)
b. Circuit rating: five amps maximum per relay contact (one circuit) at 32 VDC or 110 VAC
c. Audio lines: Balanced with respect to ground
d. Keying lines: Metallic pair
e. Ambient Temperature: 0° to 50°C
f. Primary power: 115 VAC, 60 Hz, single-phase, 115 watts
g. Equipment compatibility: Nominally installed as part of the AN/FSA-52(V) and used with the AN/FSA-17. Although designed for handling of audio and keying lines, this equipment may be used in other applications within the relay contact ratings
h. Maximum heat dissipation:
   Power supply - 20 watts
   Local switching panel - seven watts
   Remote switching panel - 14 watts
   Dimmer panel - one watt

3.3.5.9.2 Remote control equipment.

3.3.5.9.2.1 Remote switching panel SB-1996/FSA-47. Remote switching panel SB-1996/FSA-47 shall contain 10 illuminated pushbutton switches. Each pair of vertically associated switches shall remotely control the transmitter and receiver of one channel through the latching relay located in the local switching panel. The pushbutton switches shall be illuminated internally by colored lights; green to indicate the main (or normal) equipment is in use on the channel associated with the switch and red to indicate the standby (or backup) equipment is in use. The light color shall indicate the status of the associated relay in switching panel SB-1995/FSA-47. The top row of five buttons shall be marked TRANSMITTER and the bottom row of five buttons shall be marked RECEIVER. The pushbutton shall be single-pole double-throw type with provisions for pencil marking of frequency, and so forth. The marking surface shall be protected by a clear surface. The button shall be illuminated by red or green colored lights depending on the status of system. The unit dimensions shall be: 88.138 mm (3.47-inch) height, 222.35 mm (8.75-inch) width, and 220.726 mm (8.69-inch) depth.

3.3.5.9.2.2 Dimmer panel SB-4792/FSA-47. Dimmer panel SB-4792/FSA-47 shall provide capability to control the brightness of pushbuttons on remote switching panel SB-1996/FSA-47. The unit shall house a variable transformer which can be adjusted by a front panel knob marked DIMMER. The unit dimensions shall be: 81.788 mm (3.22-inch) height, 222.35 mm (8.75-inch) width, and 220.726 mm (8.69-inch) depth.

3.3.5.9.2.3 Electrical equipment cabinet CY-4042/FSA-47. Electrical equipment cabinet CY-4042/FSA-47 shall have the following dimensions: 88.138 mm (3.47-inch) height, 482.6 mm (19-inch) width, and 304.8 mm (12-inch) depth. It shall be constructed to house either two remote switching panels SB-1996/FSA-47 or one remote switching panel SB-1996/FSA-47 and one panel dimmer C-4792/FSA-47 and provide the necessary plugs and wiring for interconnection of the units. Two each electrical equipment cabinet CY-4042/FSA-47 shall be supplied with the AN/FSA-47.

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3.3.5.9.3 Local control equipment. The local equipment modules shall be housed in electrical equipment cabinets CY-4040/FSA-47 and CY-4041/FSA-47. The CY-4040/FSA-47 cabinet shall house a power supply PP-3824/FSA-47 and a local switching module, switching panel SB-1995/FSA-47. The CY-4041/FSA-47 shall house two local switching modules, switching panel SB-1995/FSA-47.

3.3.5.9.3.1 Switching panel SB-1995/FSA-47. The local switching modules, switching panel SB-1995/FSA-47 with power supply PP-3824/FSA-47, shall be mounted in a space convenient to maintenance personnel. The local switches shall have an overriding control over the remoting switches located on remote switching panel SB-1996/FSA-47. Each local switching panel SB-1995/FSA-47 shall provide control for five transmitters and five receivers. Each of the five transmitter channels and five receiver channels shall have a two-section three-position switch mounted vertically on the panel. The three switch positions shall be marked MAIN - REMOTE-STANDBY. All five top switches shall be marked TRANSMITTER and five lower switches marked RECEIVER. Each switch shall have a set of two lights mounted one above the other next to the switch. The top light shall have a green lens and the bottom a red lens. There shall be a latching-type relay inside the local unit associated with each switch. The position of the relay shall be indicated by the red and green lights and controlled by the MAIN - REMOTE - STANDBY switch. With the switch in the REMOTE position, the relay shall be remotely controlled by the appropriate pushbutton switch located on remote switching panel SB-1996/FSA-47. The status of the relay shall be indicated by which light is lit on the local panel, and also indicated by a change in the color of the light that shows through the pushbutton on the remote switching panel. The switching of the relay transfers audio and control line from one equipment (main) to another equipment (standby) located at other locations. The local switch shall override the remote switches to permit maintenance personnel to safely work on the equipment being controlled without experiencing an undesired and hazardous transfer caused by operating personnel at the remote switches. The unit shall be marked with an appropriate name plate. The unit dimensions shall be: 132.588 mm (5.22-inch) height, 222.35 mm (8.75-inch) width, and 220.726 mm (8.69-inch) depth.

3.3.5.9.3.2 Power supply PP-3824/FSA-47. Power supply PP-3824/FSA-47 supplies power for the complete system, switch control circuits, indicator lights, and latching relay. The unit shall be designed to operate on 115-VAC, 60-Hz, one-ampere power. Both legs of the input power shall be interrupted by a panel-mounted on-off switch (marked POWER ON - OFF) and be protected by two front panel-mounted 1-ampere fuxes (marked 1A). Power on shall be indicated by a panel-mounted light with green lens connected from the rectified output of the power transformer secondary. The power transformer shall have input taps for 108, 120, and 132 VAC, 50/60 Hz. Output shall be 42 Volts with center tap, four amperes. Full wave rectification shall be accomplished using two silicon diode semiconductors to provide a plus 20 VDC output; a 20 VAC output shall also be required.

3.3.5.9.3.3 Electrical equipment cabinet CY-4040/FSA-47. Electrical equipment cabinet CY-4040/FSA-47 shall house and provide the electrical connection to one each power supply PP-3824/FSA-47 and one each, local switching panel SB-1995/FSA-47. Cabinet dimensions are: 132.588 mm (5.22-inch) height, 482.6 mm (19-inch) width, and 400.05 mm (15.75-inch) depth.

3.3.5.9.3.4 Electrical equipment cabinet CY-4041/FSA-47. Electrical equipment cabinet CY-4041/FSA-47 shall house and provide the electrical connection to local switching panels SB-1995/FSA-47. Cabinet dimensions are: 132.588 mm (5.22-inch) height, 482.6 mm (19-inch) width, and 400.05 mm (15.75-inch) depth.

3.3.5.9.4 Operating spare plug-in units. One of each type plug-in unit shall be provided for operating spares.

3.3.5.10 Landline technical parameters. The AN/FSA-58(V) shall fully interface with existing leased voice landline systems in accordance with the parameters (for voice) specified in the Telephone Sets paragraph of MIL-STD-188-346.

3.3.5.10.1 Landline interface panel. This panel shall provide the necessary hybrid transformers, balance networks, and talkback bridges for ten landlines, five two-wire landlines, and five four-wire landlines. This panel shall permit voice signalling landlines to be directly controlled by the AN/FSA-58(V) radiophone circuitry.

3.3.5.11 Front panel dimensions. In the control central subgroup all front panel height and width dimensions for the standard 19-Inch Equipment Rack panel spaces shall be in accordance with FIGURE 3.

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3.3.6 Support and diagnostic subgroup. The support and diagnostic subgroup shall contain all the special test sets, special tools, and materials necessary for installation and maintenance of the AN/FSA-58(V). This subgroup shall as a minimum contain the items listed in 3.3.1.5 and shall be as specified in 3.3.6.1 through 3.3.6.6. Cables for interconnection of the equipment's subgroups are Government furnished at time of installation in a Government activity.

3.3.6.1 Special test set and adapters. The special test set and adapters of the AN/FSA-58(V) shall be packaged to provide voltages and circuitry necessary to check the performance of each plug-in module and unit in the AN/FSA-58(V). Extender cords shall be provided for servicing plug-in circuit boards. A self-contained power supply, operating from 115 VAC plus or minus 10 percent, 50 to 60 Hz plus or minus five percent, shall furnish the power necessary to operate the test set and the units under test. The primary side of the power supply shall be fused. Test set plug-in adapters shall be provided to allow testing of all plug-in modules. The adapters shall be keyed to permit only the proper connection.

3.3.6.2 Audio test set, TS-1505A/FSA-17. The audio test set, TS-1505A/FSA-17 shall be as follows:

	DESCRIPTION	REQUIREMENT
1.	Maximum audio output level into 50 Ohms	At least -21 dBm
2.	Audio output level into 50 Ohms with test set level control adjusted for 0 VU indication on test set meter. Switch in up position	-30 dBm ±2 dB
3.	Audio input level for 0VU indication on test set meter. Test set switch in middle position	-28 dBm ± dB
4.	Audio input level for 0 VU indication on test set meter. Test set switch in down position	-38 dBm ±2 dB
5.	Frequency	1000 Hertz
6.	Distortion	Five percent maximum

A diode bridge shall be added to the output of the audio test set so that the plug does not have to be polarized.

3.3.6.3 Test set TS-1664/FSA-32. Test set TS-1664/FSA-32 shall consist of a power supply and metering circuits with appropriate switches and multipliers for testing of units of AN/FSA-32 amplifier-mixer-meter group. Test set TS-1664 shall include three test adapters to provide for the connection between the test set and the unit removed from the AN/FSA-32 amplifier-mixer-muter group for testing. The test adapters shall provide connection from the appropriate plug on the test set to power supply PP-2947/FSA-32, amplifier mixer AM-2984/FSA-32, or amplifier, audio frequency AM-2933/FSA-32. A compartment for storage of the test adapters shall be provided in the lower portion of the test set. Input power to the test set shall be controlled by a front panel-mounted two-pole single-throw switch marked ON-OFF. The power cable and plug shall be of the 3-wire grounding type. Both power input lines shall be protected with 1-ampere fuses mounted on the front panel and marked LINE - 1 AMP. A front panel green lens pilot light shall be connected to the power transformer output to indicate power on. Voltage output of the power

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supply section shall be controllable from a front panel control (marked LINE MAX) adjusting a cariable resistor in the primary side of the power transformer. The output of the power supply shall be protected by two front panel mounted fuses; one 1.5-ampere fuse in the output and one 0.25-ampere fuse in the minus 10-Volt output. The unit also produces a plus 5-Volt output. A front panel amber lens pilot light shall be connected to indicate the presence of low voltage. The two fuses and pilot lights shall be marked LOW VOLTAGE and 1.5 AMP and 0.25 AMP. The plus 5-Volt output shall be (bias) adjustable thru a front panel variable resistor marked BIAS MAX. The line and bias voltages shall be connected to the VU meter by the BIAS LINE ADJUST switch (two-pole three-position spring-loaded both ways). It shall be marked #ADJUST FOR OVU$. When the switch is held to the left (LINE) position, the VU meter shall be connected to monitor the adjustment of the LINE control. The sensitivity of the circuit shall be such that a zero reading on the VU meter means that the line input is properly adjusted. When the center normal switch is held to the right shall permit adjustment of the BIAS control to have the meter read zero VU for properly adjusted bias. In the center (at rest) position, the VU meter input sensitivity shall be controlled by the front panel-mounted meter switch marked METER. The setting of the METER switch shall be marked +5V, -10V, -12V, INPUT, and OUTPUT. The front panel mounted FUNCTION switch shall connect the metering capability of the test set to the appropriate plug mounted on the front panel of the test set. These plugs shall mate with the test adapter supplied with the test set. One marking AMPLIFIER - POWER SUPPLY - MIXER for each plug and function shall be permitted. A front panel-mounted two-pole three-position spring-loaded both ways center-rest switch shall be provided. This switch shall be marked #LO - HI - MUTE$. In the center position, no ground shall be connected to the unit under test and the gain of the amplifier shall be normal. When the switch is held to the left (LO), it shall ground out the HI-LO line to the unit under test, reducing its output gain by 15 dB. When the switch is held to the right (MUTE), it shall provide a ground on the mute line to the unit under test and shall reduce the gain of the amplifier by 40 dB. A second section of this switch shall change the meter multipliers as necessary for a proper reading on the VU meter. Injecting an audio signal into a unit under test shall be accomplished by this unit. The signal generator (not supplied) shall be connected to a binder post unit (one black and one red) mounted on the front panel of the test set. This input shall be marked SIGNAL GENERATOR. The signal shall be routed fron this input to a front panel-mounted two-section five-position rotary switch marked CHANNEL 1, 2, 3, 4, and 5. This switch shall connect the test audio signal and the mute signal to the appropriate channel in the unit under test. The test set TS-1664/FSA-32 dimensions shall be 215.9 mm (8.50-inch) height, 368.3 mm (14.50-inch) width, and 177.8 mm (7-inch) depth.

3.3.6.4 Riveting tool set. Two riveting tool sets shall be provided and contain: riveting tools, de-riveting tool, and twice the quantity of rivets necessary to attach mounting bracket described in 3.3.3.16.5. FIGURE 6 contains fabrication details for the riveting and deriveting tools.

3.3.6.5 Wirewrap tool set. Two wirewrap tool sets shall be provided to wrap and unwrap No. 22 and No. 19 guage wire to the 0.045-inch square terminal posts, Gardner Denver Co., or equivalent.

3.3.6.6 Extender cards. Extender cards shall be provided for servicing all printed circuit (p.c.) boards in the AN/FSA-58(V).

3.3.7 AN/FSA-58 amplifier characteristics. The amplifier characteristics specified in 3.3.6.1 through 3.3.6.2 shall apply for the AN/FSA-58 except for the AN/FSA-32.

3.3.7.1 AN/FSA-58 amplifiers frequency response. The frequency response for each amplifier in the AN/FSA-58(V) shall be ±1 dB from 300 Hz to 3000 Hz from the base frequency of 1000 Hz and shall be down at least 20 dB at 100 Hz with continuous roll-off below 100 Hz. High frequency roll-off above 3000 Hz shall be at least 3 dB per octave.

3.3.7.2 AN/FSA-58 amplifiers input impedances. The specified input impedances for each amplifier in the AN/FSA-58(V) system shall be maintained within ±10 percent of the specified value over the 300 Hz to 3000 Hz frequency range.

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3.3.7.3 AN/FSA-58(V) amplifier distortion. Distortion for each amplifier in the AN/FSA-58(V) shall be no more than three percent at maximum specified output level.

3.3.8 AN/FSA-58(V) component isolation. The equipment shall be designed and connected with proper isolation so that failure or removal or insertion of a plug-in unit, assembly, amplifier, power supply, test unit, or part shall not interact to cause other portions of the equipment to be put out of operation or adversely affected. Transient clicks or momentary disruptions while removing or inserting the aforementioned items are unacceptable.

3.3.9 AN/FSA-58(V) crosstalk. Circuit design shall provide 60-dB isolation between channels and amplifier outputs. The circuit design for radiophone channels shall provide 80-dB isolation.

3.3.10 AN/FSA-58(V) connector cable assembly. Except for the power circuit, inter-connection to each of the units within a subgroup shall be accomplished with multiconductor telephone type cable No. 22 gauge wire. Care shall be taken in the dress of the cable to maintain color code pairing and bending ratios according to good engineering practice. Square terminal posts (0.045-inch) shall be used throughout the equipment.

3.3.11 Plug-in relays. Plug-in relays with dust covers and retainers shall be provided throughout AN/FSA-58(V). Relay contacts shall be bifurcated type.

3.3.12 Silicon semi-conductors. Silicon semi-conductors shall be provided throughout the AN/FSA-58(V) except where otherwise specified.

3.3.13 Dimension tolerances. Tolerances indicated on FIGURE 2 shall be maintained for all modules and front panel mounting brackets for the control position modules and units.

3.3.14 AN/FSA-58(V) number logic. A simple number logic shall be incorporated for associating units and circuits with channels and operator control positions to permit efficient installation and maintenance.

3.3.15 Parts location. Parts associated with a specific assembly, subassembly, unit, or module shall be located in that assembly, subassembly, unit, or module unless specifically approved otherwise by the procuring activity.

3.4 Operating plug-in spares. For each equipment supplied, one plug-in module, unit, and subassembly of each type used in the equipment shall be provided as an operating spare. A complete rectifier/charger shall be provided as an operating spare.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; b) Government product inspection to measure quality of the product to be offered for acceptance; c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The methods of examination and testing of the equipment shall be classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
    1. Production inspection (see 4.4.1)
    2. Production control inspection (see 4.4.2)
c. Reliability demonstration (see 4.5)
d. Maintainability demonstration (see 4.6)
e. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.2) two equipments shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements specified herein. First article inspection shall include the tests specified in TABLE III.

4.3.1 Electromagnetic compatibility tests. The equipment shall be tested to verify the requirements of 3.2.13 using the test methods of MIL-STD-462.

4.3.2 Environmental testing. For tests requiring a chamber, the equipment shall be divided into its subgroups for purposes of environmetnal tests, subject to the approval of the procuring activity.

4.3.3 Rough handling tests. Rough handling tests shall be conducted in accordance with MIL-P-116.

4.3.4 Design and construction test. A complete equipment operating test (see 4.4.1.1) shall be conducted on the AN/FSA-58(V). The reduced performance under conditions of extreme voltages specified in 3.2.14 shall be documented.

4.3.5 Reliability testing. Reliability testing shall be performed on a basic equipment (3.3.1) which has been subjected only to production inspection. Testing shall be performed at Test Level E with Test Plan XXVIII of MIL-STD-781. The basic equipment shall be subjected to 2000 operating hours at a temperature in accordance with 3.2.1.

4.3.6 Leakage current test. Method of measurement of current leakage shall be in accordance with UL 492, paragraphs 195 through 200.

4.3.7 Human factors engineering testing. Human factor engineering tests shall consist of qualitative observations during the a) design and construction test (equipment operating test), and b) the maintainability demonstration to determine compliance with the human factors requirements specified in 3.2.12.

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TABLE III. First article examination and tests.

Examinations or tests	MIL-E-16400 requirement paragraph	MIL-E-16400 test paragraph	Requirement paragraph (MIL-C-28771)	Test paragraph (MIL-C-28771)
Surface examination	Surface examination	Surface examination		
Weight	Weight	Weight		
Size	Size	Size		
Parts and materials	Parts, materials and processes	Parts and materials		4.4.1
Finish	Painting	Finish		4.4.1
Marking	Marking	Marking		4.4.1
Safety	Safety criteria	Safety	3.2.12	4.4.1
Workmanship	Workmanship	Workmanship		4.4.1
Pre-performance		Pre-performance test		
Design and construction	Electrical and mechanical design and construction		3.3	4.3.4
	Temperature test sequence		3.2.1	
Temperature	Operating temperature ranges, Non-operating temperature ranges			
Low temperature		Low temperature	3.2.1	4.4.2
High temperature	Operating temperature ranges, Non-operating temperature ranges	High temperature	3.2.1	4.4.2
Humidity	Humidity	Humidity		4.4.2
Magnetic	Magnetic characteristics	Magnetic test		
Weld	Welding, structural	Weld test		
Steady state voltage and frequency	Steady state voltage and frequency	Steady state voltage and frequency		
Transient voltage	Electrical power characteristics	Transient voltage		
Transient frequency	Electrical power characteristics	Transient frequency		
Power interruption	Electrical power characteristics	Power interruption		
Power and power factor	Electrical power characteristics	Power and power factor		
Insulation resistance	Insulation resistance	Insulation resistance		4.3.6
Leakage current	Leakage current (equipment)	Leakage current	3.2.13	4.3.1
Electromagnetic compatibility			3.2.3	4.3.3
Rough handling		Thermal test		
Thermal design	Thermal design	Reliability test and and demonstration	3.2.4	4.3.5
Reliability testing	Reliability requirement		3.2.5	4.6
Maintainability demonstration			3.2.12	4.3.7
Human factors	Human engineering			

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4.4 Quality conformance inspections.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. Functional tests and examinations shall be conducted on the items listed in 3.3.1, to evaluate and determine the acceptability of workmanship and fabrication techniques used in the assembly process. Tests and examinations shall be performed which will isolate defective items and measure attributes which cannot be inspected at the end item level and shall include the following:

a. Workmanship during fabrication of all items (Requirement 9 of MIL-STD-454)
b. Examination for marking and safety
c. Examination of coating during fabrication including coatings on printed wiring assemblies
d. Correct operation of all controls, indicators, switches, and adjustments
e. Subassembly tests to determine correct operation of printed wiring assemblies and modules
f. Tests for all inputs and outputs for each operating mode not covered by a specific test requirement
g. Tests for correct operation of forced air cooling system
h. Examination and test of extender boards and cables when these are not used in higher level tests
i. Examination of parts and materials
j. Examination for ease of operation of mechanical assemblies

4.4.1.1 Equipment operating test. The equipment, assembled to include all parts required for delivery of end items, shall be interconnected to simulate an operating test and shall be subjected to an operating test to demonstrate conformance with the requiremetns of Section 3.

4.4.1.1.1 Sampling for equipment operating test. One equipment from each successive eight equipments produced, or one every two months, if less than eight equipments are produced in two months,shall be selected for the equipment operating test. This initial equipment operating test shall be conducted on the first production offered for delivery. The Government shall select the particular configuration of the sample to be tested.

4.4.2 Production control inspection. Production control inspection shall be performed on the equipment and shall consist of the tests specified in TABLE IV.

TABLE IV. Production control inspection.

Inspection	Reference
Low temperature	MIL-E-16400
High temperature	MIL-E-16400
Humidity	MIL-E-16400

4.4.2.1 Sampling for production control inspection. One sample shall be selected from each 50 equipments, or fraction thereof, produced. The first sample shall be selected from the first month's production.

4.4.2.2 Nonconforming sample units. If a sample unit fails an inspection, the contractor shall immediately investigate the cause of failure and shall report to the Quality Assurance Representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the Quality Assurance Representative does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.2.3 Reinspection of conforming production control units. Unless otherwise specified, sample units which have been subjected to and passed production control inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

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4.5 Reliability demonstration. Reliability demonstration shall be performed in accordance with MIL-STD-781.

4.5.1 Reliability qualification tests. Qualification testing shall be performed on at least two production basic equipments (3.3.1). Testing shall be performed at Test Level E with Test Plan IV, and shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for Test Plan IV. Temperature shall be as specified in 3.2.1.

4.5.2 Rejected equipment. When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer designs, corrective action shall be performed on all equipment delivered under the contract.

4.5.3 Failure. For definition of failure see 6.7.

4.5.4 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning as required and shall be subjected to the production tests of 4.4.1 to ensure that it conforms to the requirements specified herein. After reconditioning, reinspection, and retesting, the equipment may be delivered as part of the contract quantity provided it meets all terms and conditions of the contract for acceptance.

4.6 Maintainability demonstration. Maintainability demonstration shall be performed in accordance with MIL-STD-471. Accept/reject criteria shall be in accordance with Method 4 of MIL-STD-471. For maintainability demonstration purposes the equipment repair time (ERT) is equivalent to the median-corrective-maintenance-time.

4.7 Equipment burn-in. All deliverable equipments shall be subjected to a temperature burn-in test with the environment temperature equal to or greater than 50°C, +0°C -5°C, with the equipment operating in the worst case mode for a cumulative operating time of 60 hours with the last 30 hours failure free. Equipment monitoring shall include as a minimum the functional examinations to ensure proper operation under the worst case operating mode. A performance examination for proper operation, a minimum of twice every 24 hours and just prior to end of burn-in, shall be performed during this test. No adjustments of the equipment, other than normal operating mode changes, shall be permitted during the burn-in. The burn-in test shall be performed prior to production inspection.

4.8 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance to the requirements of Section 5.

5.0 PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6.0 NOTES

6.1 Intended use. The equipment is intended for use in providing air traffic control and monitor operations at various Naval shore sites.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. Levels of preservation packaging, packing, and marking (see 5.1)
d. When vertical mounting is required (3.3.3.1)

6.3 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data generated by this specification include, but are not restricted to, the following:

a. Reliability program plan
b. Reliability prediction
c. Reliability reports
d. Maintainability program plan
e. Maintainability prediction
f. Test point data
g. General purpose test equipment
h. Equipment requiring human engineering
i. Color illumination details
j. Quality plan
k. Test procedures
l. Test agenda program
m. First article test report
n. Reliability demonstration test plans
o. Reliability test report
p. Maintainability test report
q. Approval of non-listed test equipment
r. Electromagnetic interference control
s. Electromagnetic interference control plan
t. EMI/EMC test plan
u. EMC test report

6.4 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 Initial shipment. Initial shipment of end items will be contingent upon successful completion of the Reliability Qualification Test.

6.6 Production control inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. Initial shipment of end items will be contingent upon successful completion of porduction control inspection.

6.7 Definitions. Definitions of terms used in this specification are as follows:

ASSEMBLY	Assembly is defined as a number of parts of sub-assemblies or any combination thereof, joined together to perform a specific function and capable of disassembly (such as Audio Oscillator, Amplifier, Synthesizer, Crystal Oscillator)
COMMUNICATIONS CONTROL EQUIPMENT	The complete apparatus and circuitry necessary to provide flexible communications operations and monitoring in aircraft traffic control and surveillance operations

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CONTROL CENTRAL	Control central is defined as all relays, detector, amplifiers, selectors, indicators, and other electronic, electrical, or electromechanical equipment directly effected by operation at the operator and supervisor control sub-groups.
EQUIPMENT	Equipment is defined as all possible combinations of units required to perform a function. For this procurement, AN/FSA-58(V) is the equipment
FAILURE	Failure is defined as any malfunction or parameter deviation that prevents the equipment from performing full capability in all modes, in the specified operational environment, to operating parameter levels set forth herein
INSTANTANEOUS INTERPHONE OVERRIDE RESPONSE (IIOR)	Instantaneous interphone override response (IIOR) is defined as the capability of an operator to reply to an interphone override call with performing any manual control operation
INTERPHONE DIRECT ACCESS	Interphone direct access is defined as direct interphone communications access, without any lockout to an operator control subgroup, used in the private interphone channel communication.
INTERPHONE OVERRIDE (AUDIO MIX)	Interphone override (audio mix) is defined as the capability of a calling operator to select and impose a communication directly into the selected operator(s) or supervisor headphone or speaker
KEY LOCKOUT	Key to lockout is defined as obtaining positive control of a radiophone transmitter by transmit key switch operation, which excludes control and modulation to all other operators while the key switch is operated
LANDLINE CHANNEL	Landline channel is defined as two-way communications path with various signalling on telephone private lines, Government-owned or leased from telephone companies
NORMAL INTERPHONE CALL	Normal interphone call is defined as the calling operators receipt of a visual answering indication before speaking
OFF-LINE	Off-line is defined as tests performed while the tested equipment is not performing its normal operational function
ON-LINE	On-line is defined as a test which is performed while the equipment is in normal operational use of the supported equipment
OPERATOR CONTROL SUBGROUP	The assembly, switches, indicators, and other manual or visual devices used by the operator in directing and monitoring air traffic operations by radiophone, or interphone, and landline (telephone private line) communications

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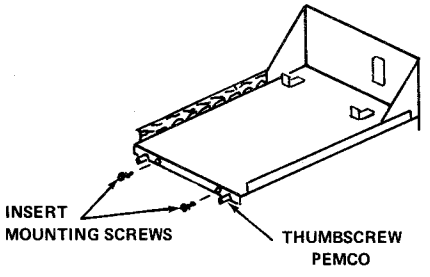
PRIVATE INTERPHONE CHANNEL	Private interphone channel is defined as direct inter-communications path between two operator control subgroups without interference to any other subgroups
RADIOPHONE CHANNEL	Radiophone channel is defined as a communications path for sending and receiving voice signal on a radio frequency carrier
RADIOPHONE OVERRIDE	Radiophone override is defined as seizing control of a radiophone transmitter by the supervisor at the supervisor control subgroup
SELECT TO LOCKOUT	Select to lockout is defined as obtaining positive control of a radiophone transmitter by the RECEIVE-TRANSMIT switch selection which excludes control and modulation to all other operators until deliberately released
SIDETONE	Sidetone is defined as present in the headphones when an operator is connected to interphone and landline channels in any combination. The absence of sidetone indicates the existence of a fault in a communications channel
STANDBY CHANNEL	Standby channel is defined as a set of components connected together, line adjusted, and tuned to a radiophone frequency, forming a radiophone channel of communications that can be instantaneously substituted, by operator or maintenance technicians, for the set of equipment and lines normally used on that frequency
SUBASSEMBLY	Subassembly is defined as two or more parts which form a portion of an assembly, but having parts which are individually relaceable (such as modules and printed circuit boards with components)
SUBGROUP MONITOR	Subgroup monitor enables monitoring all voice communications (radio phone, interphone, and landline channels) of an operator control subgroup
SUBGROUP RECORDING	Subgroup recording is defined as the recording of all audio from headphone, speaker, and microphone amplifier outputs from an operator control subgroup on one recorder channel
SUPERVISOR CONTROL SUBGROUP	Supervisor control subgroup is defined as the assembly of controls, displays, and other manual or visual devices required for an operator to perform a supervisory function over the communications control equipment

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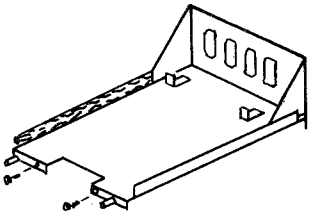
TRUE SIDETONE	True sidetone is defined as return of an operator's voice to his headphone via radiophone transmitter and receiver circuits. This sidetone is 12 dB below the level of signals received simultaneously in the headphones from radiophone channels not modulated by the operator. The absence of this sidetone indicates a failure in the radiophone channel communications
UNIT	Unit is defined as an assembly or any combination of parts, subassemblies and assemblies mounted together, normally capable of independent operation in a variety of situations. For this procurement, the units are AN/FSA-32, AN/FSA-47(V), and so forth

Preparing activity:
NAVY-EC
(Project 5895-N58)

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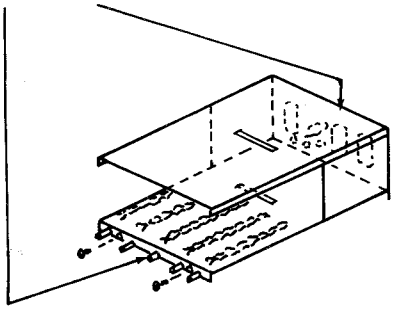


SPEAKER AMPLIFIER INSERT (250-36)



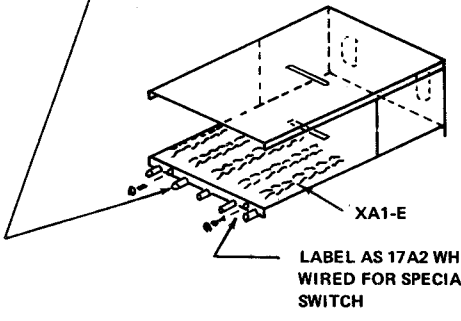
CONSOLE AMPLIFIER INSERT (250-35)

LABEL AS 16A1 OR 16A2
HERE IN ACCORDANCE WITH CABLE
CONNECTED AT J2 AND JUMPER
CONNECTED AT E1, E2, E3



RDPH & INPH SWITCH INSERT (250-37)

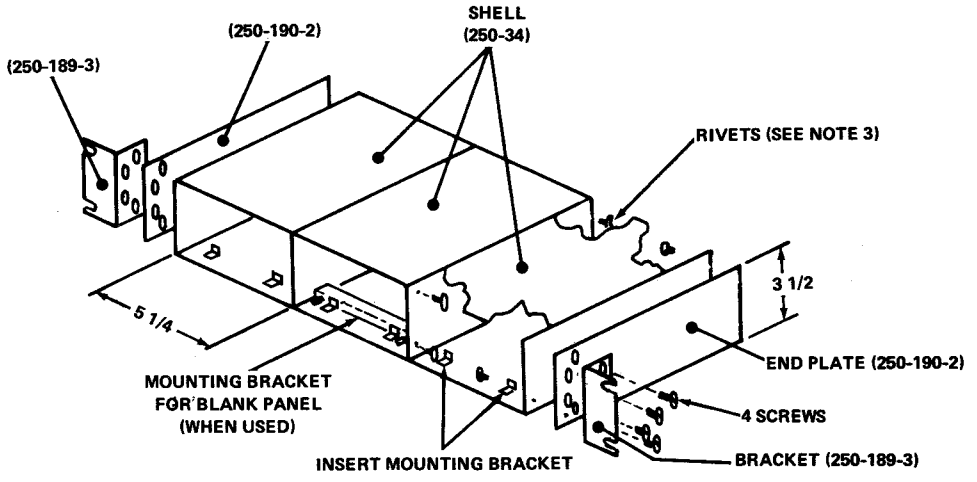
LABEL AS 17A1 FOR "T" AND "C" SWITCH FUNCTION



SUPERVISOR'S STATUS AND SPECIAL SWITCH INSERT (250-38)

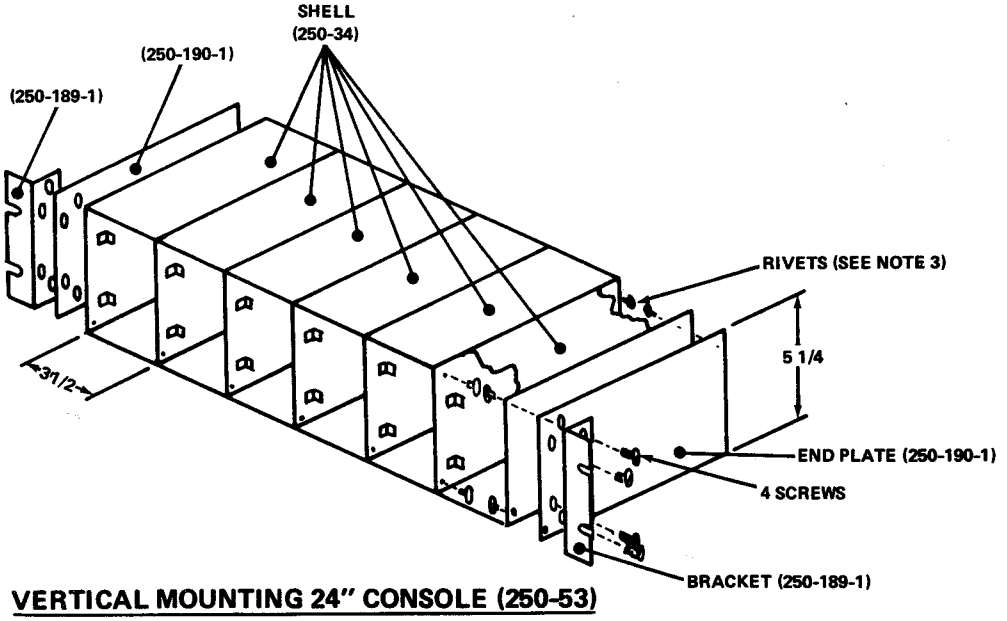
FIGURE 1. Outline and mounting dimensional data for AN/FSA-58(V) (sheet 1 of 3)

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HORIZONTAL MOUNTING 19" RACK CONSOLE (250-50)

FOR 24" CONSOLE MOUNTING (250-51), ADD ONE SHELL (250-34)



VERTICAL MOUNTING 24" CONSOLE (250-53)

FOR 19" CONSOLE MOUNTING (250-52) REMOVE ONE SHELL (250-34)

FIGURE 1. Outline and mounting dimensional data for AN/FSA-58(V) (sheet 2 of 3).

NOTES:
1. DIMENSIONS ARE IN INCHES.
2. FOR PARTS LISTING DETAIL, SEE SUMMARY LIST OF INSTALLATION MATERIAL, DIV G RW6D901.
3. USE RIVETING TOOL (250-55) FOR RIVETING SHELLS AND END PLATES. DE-RIVETING TOOL IS PART 250-56.

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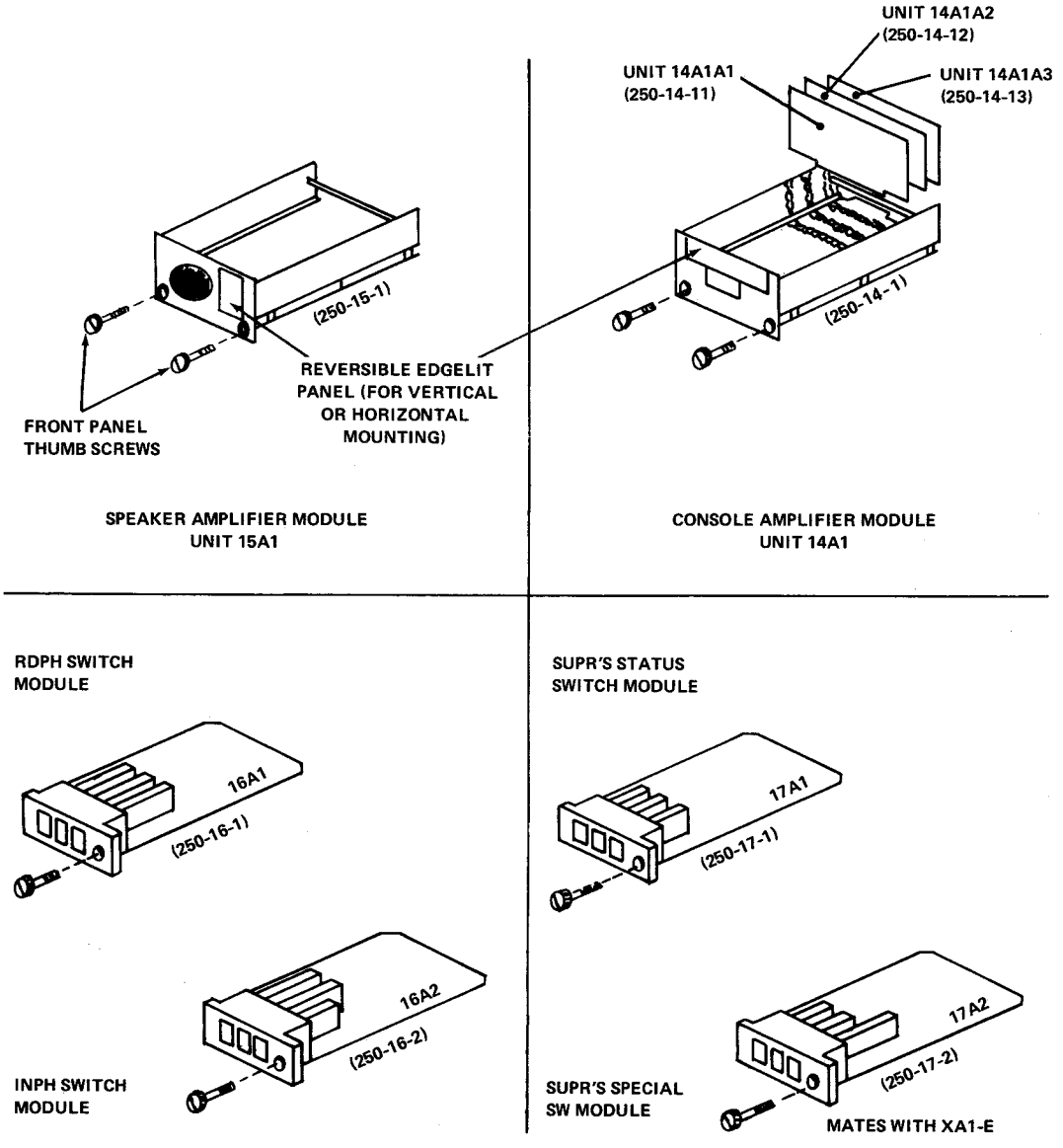
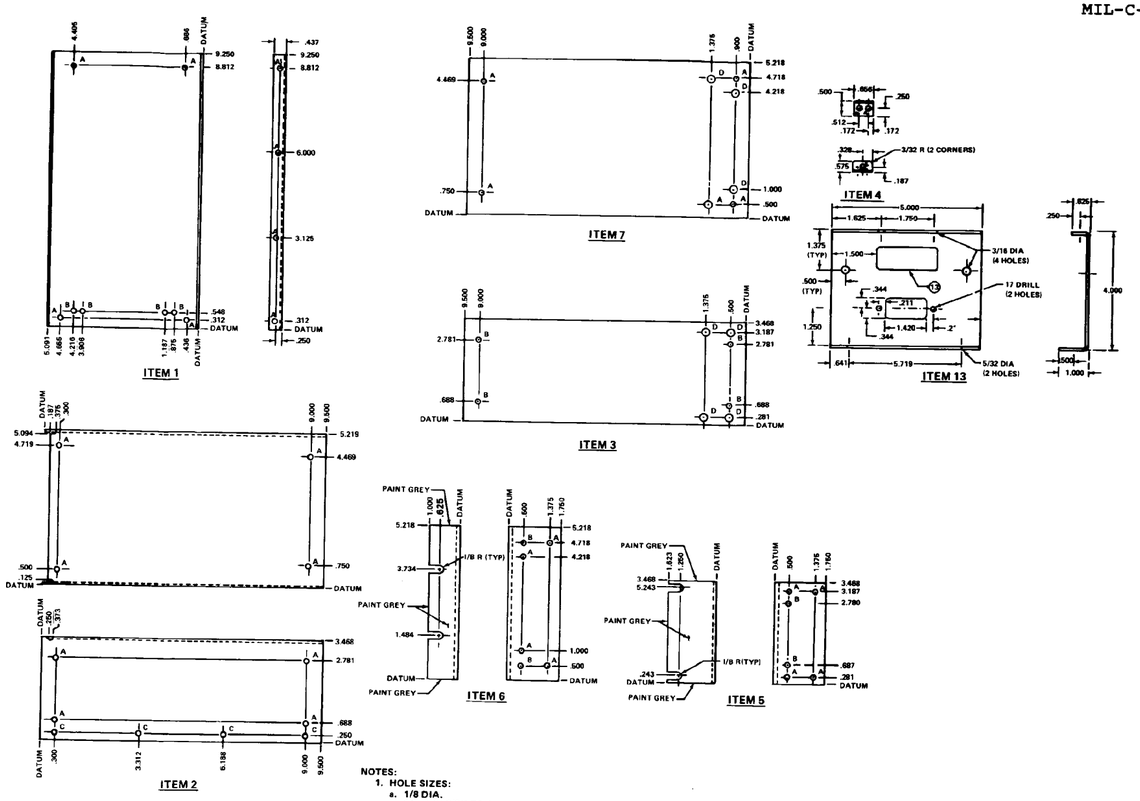


FIGURE 1. Outline and mounting dimensional data for AN/FSA-58(V) (sheet 3 of 3).

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NOTES:

1. HOLE SIZES:
   a. 1/8 DIA.
   b. 1/8 DIA. C'SK TO 0.235 DIA. FAR SIDE
   c. 1/8 DIA. C'SK TO 0.235 DIA. NEAR SIDE
   d. NO. 28 DRILLS C'SK FOR NO. 6-32 PEM STUD
   e. NO. 27 DRILL

2. MATERIAL:
   ITEMS 1, 2, 3, 4, AND 7 ALUM ALLOY
   ITEM 5, 1-3/4 x 1-3/4 x 1/8 ALUM ANGLES
   ITEM 6, 1 x 1-3/4 x 1/8 ALUM ANGLE
   ITEM 13, 0.094 ALUM ALLOY

3. FINISH
   ITEMS 1, 2, 3, 4, AND 7, ETCH AND GOLD IRIDITE
   ITEMS 5 AND 6, ETCH AND GOLD IRIDITE PRINT WITH GRAY IAW MIL-P-15934 AS INDICATED
   ITEM 13, ETCH AND GOLD IRIDITE

4. TOLERANCES: FRACTIONS, ± 1/24, DECIMALS, ± 0.005

FIGURE 2. Console cases assembly and fabrication details, (sheet 1 of 2).

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LIST OF MATERIAL				
ITEM	QTY	DESCRIPTION	TYPE OR STOCK NO.	REMARKS
1	3	BOTTOM PLATE		SH 2
2	3	TOP PLATE		SH 2
3	2	SIDE PLATE		SH 2
4	6	BRACKET		SH 2
5	2	MOUNTING BRACKET		SH 2
6		MOUNTING BRACKET	USED ON VERTICAL ASSEMBLY ONLY	SH 2
7		TOP PLATE	USED ON VERTICAL ASSEMBLY ONLY	SH 2
8	8	NUT, HEX	NO. 6-32, UNC	
9	8	WASHER, LOCK	NO. 6 CAD, PLATE	
10	8	PEM STUD	NO. 6 UNC - 24 x 5/16	FH6-325
11	16	RIVET, C'SK HEAD	1/8 DIA. ALUM.	
12	24	RIVET, PAN HEAD	1/8 DIA. ALUM.	
13	1	NAME PLATE	FAB (SEE RHA-6-5-41060)	

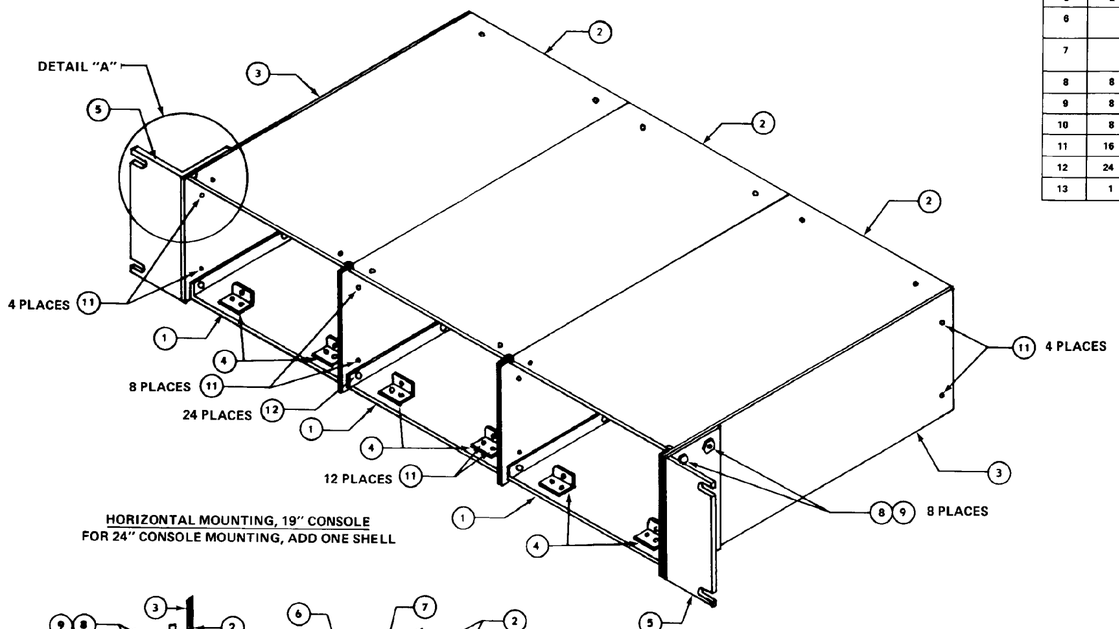
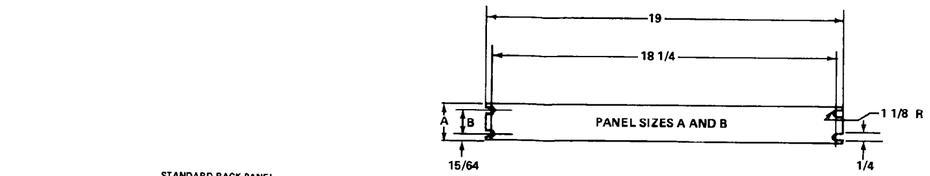


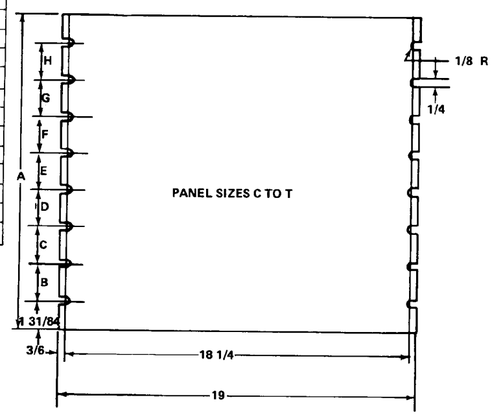
FIGURE 2. Console cases assembly and fabrication details (sheet 2 of 2).

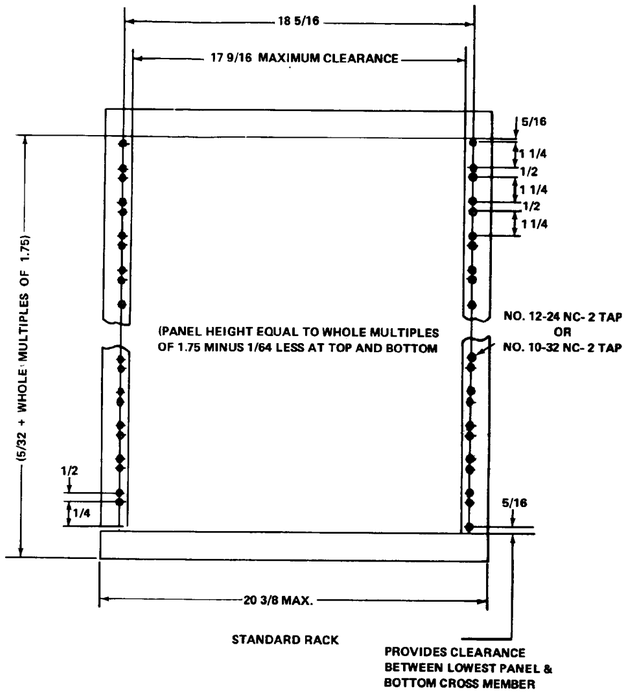
Source: https://assist.dla.mil -- Downloaded: 202610-20T15:04Z
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STANDARD RACK PANEL

PANEL SIZE	A	TABLE OF ST'D PANEL DIMENSIONS B	C	D	E	F	G	H
A	1 23/32	1 1/4						
B	3 15/32	3						
C	5 7/32	2 1/4						
D	6 31/32	4						
E	8 23/32	1 3/4	2 1/4	1 3/4				
F	10 15/32	2 1/4	3	2 1/4				
G	12 7/32	1 3/4	5 3/4	1 3/4				
H	13 31/32	3 1/2	4	3 1/2				
J	15 23/32	3 1/2	5 3/4	3 1/2				
K	17 15/32	3 1/2	7 1/2	3 1/2				
L	19 7/32	5 1/4	5 3/4	5 1/4				
M	20 31/32	1 3/4	5 1/4	4	5 1/4	1 3/4		
N	22 23/32	5 1/4	1 3/4	5 3/4	1 3/4	5 1/4		
P	24 15/32	1 3/4	5 1/4	7 1/2	5 1/4	1 3/4		
Q	26 7/32	1 3/4	7	5 3/4	7	1 3/4		
R	27 31/32	1 3/4	5 1/4	3 1/2	4	3 1/2	5 1/4	1 3/4
S	29 23/32	1 3/4	5 1/4	3 1/2	5 3/4	3 1/2	5 1/4	1 3/4
T	31 15/32	1 3/4	5 1/4	3 1/2	7 1/2	3 1/2	5 1/4	1 3/4







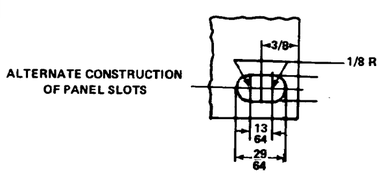
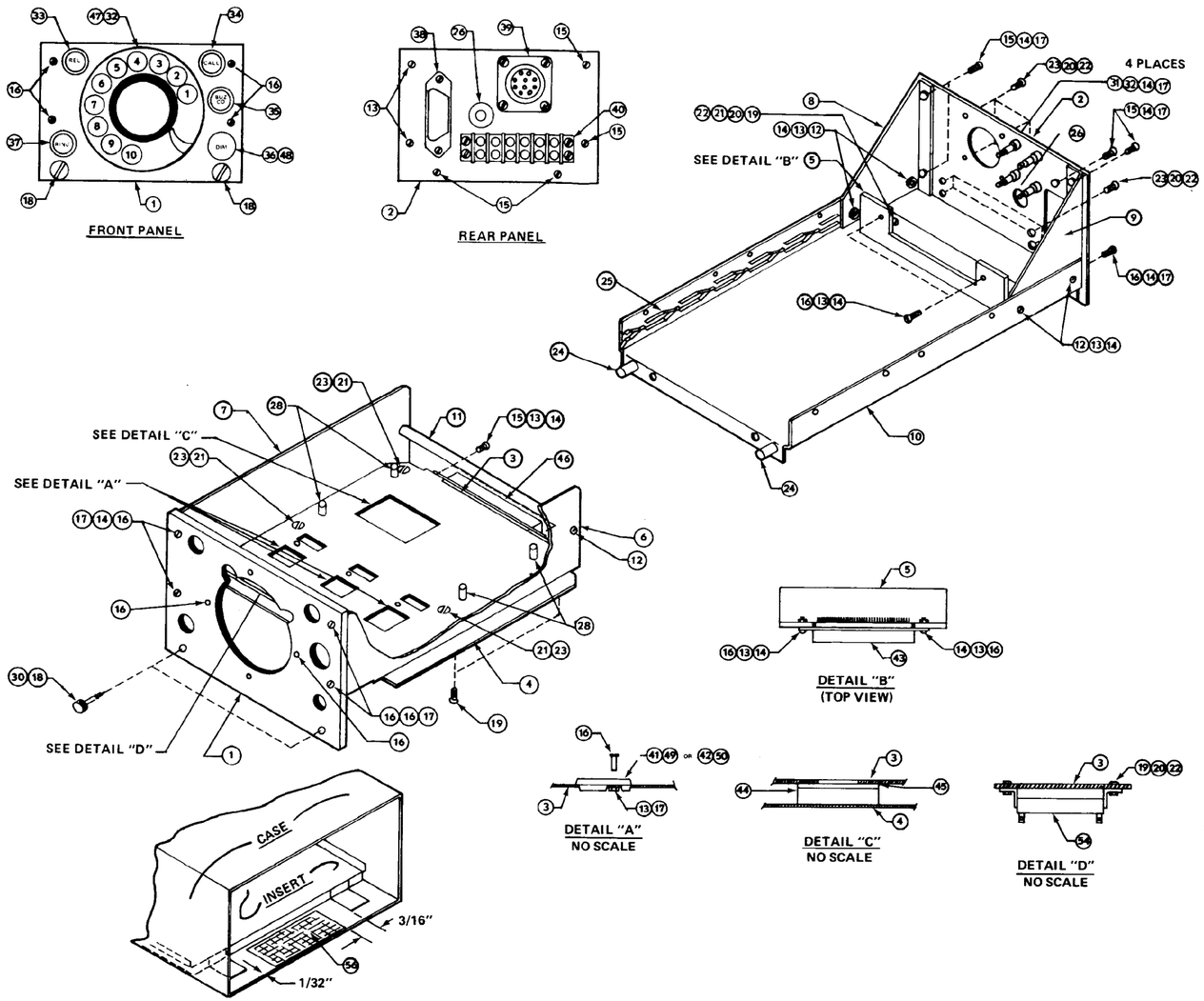


FIGURE 3. Standard 19-inch relay rack and panel dimensions.

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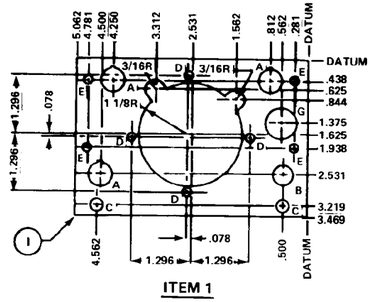


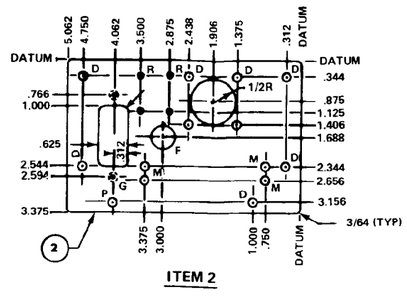
LIST OF MATERIAL

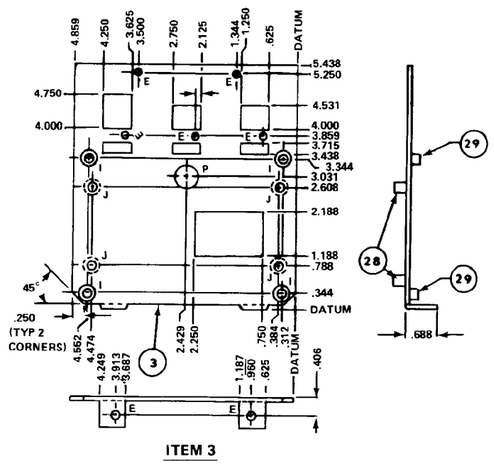
ITEM	QTY	DESCRIPTION	TYPE OR STOCK NUMBER	REMARKS
1	1	PANEL, FRONT		
2	1	PANEL, REAR		
3	1	CHASSIS, FRONT UNIT		
4	1	PLATE, BOTTOM		
5	1	BRACKET, CONNECTOR		
6	1	PLATE, SIDE		
7	1	PLATE, SIDE		
8	1	BRACKET, CHASSIS		
9	1	BRACKET, CHASSIS		
10	1	CHASSIS, REAR UNIT		
11	1	BRACE, FRONT UNIT	1/4" DIA ALUM ROD	
12	6	SCREW, PHILLIPS FL HD	NO. 4-40 x 1/4 CAD. PLATE	
13	11	NUT, HEX	NO. 4-40 CAD. PLATE	
14	18	WASHER, LOCK	NO. 4 CAD. PLATE	
15	12	SCREW, PAN HEAD	NO. 4-40 x 5/16 CAD. PLATE	
16	11	SCREW, PAN HEAD	NO. 4-40 x 5/8 CAD. PLATE	
17	13	WASHER, FLAT	NO. 4 CAD. PLATE	
18	2	SCREW, CAPTIVE	NO. 8-32 CAD. PLATE	
19	8	SCREW, PAN HEAD	NO. 8-32 x 5/16 CAD. PLATE	
20	14	NUT, HEX	NO. 8-32 CAD. PLATE	
21	8	WASHER, FLAT	NO. 6 CAD. PLATE	
22	14	WASHER, LOCK	NO. 6 CAD. PLATE	
23	12	SCREW, PAN HEAD	NO. 6-32 x 1/8 CAD. PLATE	
24	2	STANDOFF, 1/4 D. x 3/8	NO. 8-32 THD	
25	2	GUIDE, P.C. BOARD		
26	1	GROMMET, RUBBER	1/2"	
27	14	PEM FASTENER	NO. 4-40	
28	4	STANDOFF, 1/4 D. x 1/4	NO. 6-32, SMITH NO. 4118	
29	4	STANDOFF, 1/4 D. x 3/16	NO. 6-32	MODIFY ITEM NO. 28
30	8	WASHER, LOCK	NO. 6 CAD. PLATE	
31	4	STANDOFF, 1/4 D. x 3/8	NO. 6-32, SMITH NO. 4121	
32	1	DIAL	AUTOMATIC ELECTRIC NO. CH25	
33	1	SWITCH, PUSH-BUTTON	DIALCO NO. 183-A015-376	
34	1	LIGHT, INDICATOR	DIALCO NO. 183-9830-371	
35	1	SWITCH, PUSH-BUTTON	MARCO-OAK NO. 6605 (PRESS-LITE) BLUE LENS CAP	
36	1	CONTROL, DIMMER	MALLORY TYPE VW-1R	
37	1	SWITCH, PUSH-BUTTON	DIALCO NO. 183-A015-372	
38	1	CONNECTOR, RECEPTACLE	AMPHENOL NO. 57-40240	
39	1	CONNECTOR, RECEPTACLE	AMPHENOL NO. MS3102 18-15	
40	1	TERMINAL STRIP	CINCH-JONES NO. 353-16-06-001	
41	2	SOCKET, RELAY	ALLIED NO. 30053-2	
42	1	SOCKET, RELAY	ALLIED NO. 30055-26	
43	1	CONNECTOR, RECEPTACLE	AMPHENOL NO. 57-20500	
44	1	RECEIVER CAPSULE, TELE.	KELLOGG TYPE 600002-370-001	
45	1	BUSHING, RUBBER	FABRICATE FROM 1/8" D x 6" RUBBER TUBING	
46	1	CONNECTOR, RECEPTACLE	AMPHENOL NO. 57-10500	
47	1	LIGHT DIAL	DIALCO NO. MS25446-7	
48	1	KNOB	RAYTHEON TYPE 70-1-EG	
49	2	RELAY	ALLIED CONTROL CO., INC. NO. T-183-4C, 24 V	
50	1	RELAY	ALLIED CONTROL CO., INC. NO. T-183-6C, 24 V	
51	4	TERMINAL, DOUBLE TURRET	H.H. SMITH NO. 52-1500	
52	4	SCREW, PAN HEAD	NO. 4-40 x 1/4, CAD. PLATE	
53	3	SEMICONDUCTOR, DEVICE	ERIE ED3004A/1N2070	
54	1	RESISTOR, FIXED	230 OHM, 10 W OHMITE	
55	2	RESISTOR, FIXED	690 OHM, 1/4 W., 10%	
56	1	NAMEPLATE	FABRICATE, SEE RHA-6-F-41060	
57	1	CASE	FABRICATE, SEE RHA-6-F-41022	

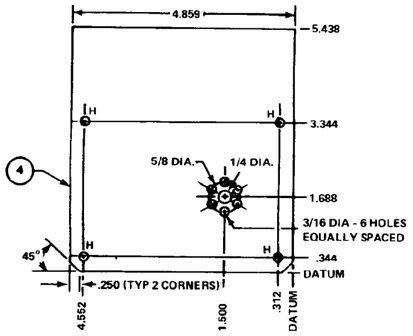
FIGURE 4. Dial unit assembly and fabrication details. (Sheet 1 of 6).

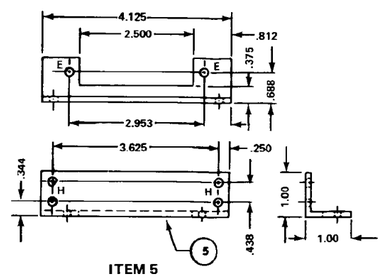
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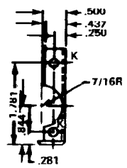


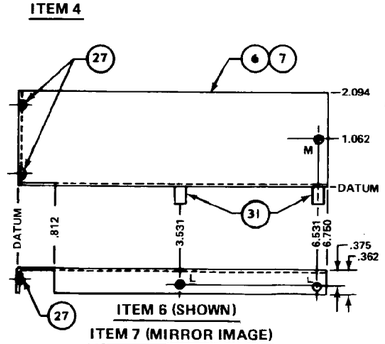


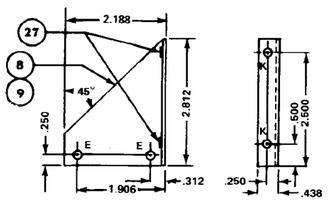


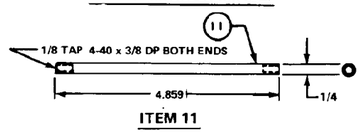


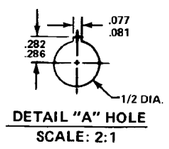
HOLE SIZE	
LETTER	SIZE
A	1/2" DIA, EXT KEY SLOT (SEE DETAIL "A" HOLE)
B	3/8" DIA
C	3/16" DIA
D	9/64" DIA
E	1/8" DIA
F	1/2" DIA
G	No. 3-48 PEM FASTENER, FAR SIDE
H	5/32" DIA
I	NO. 12 DRILL, C'BORE 5/16 x 1/32 DP, NEAR SIDE
J	NO. 12 DRILL, C'BORE 5/16 x 1/32 DP, FAR SIDE
K	NO. 19 DRILL, NO. 4-40 PEM FASTENER
L	NO. 12 DRILL
M	1/8" DIA, CS'K (NEAR SIDE) FOR F.H.M.S.
N	3/32" DIA
P	7/16" DIA
R	NO. 31 DRILL

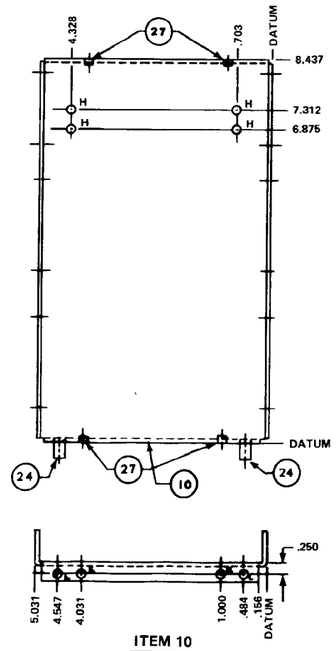


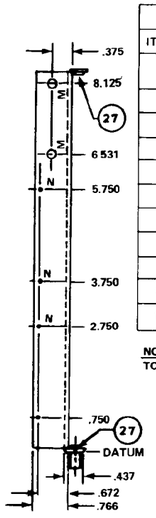










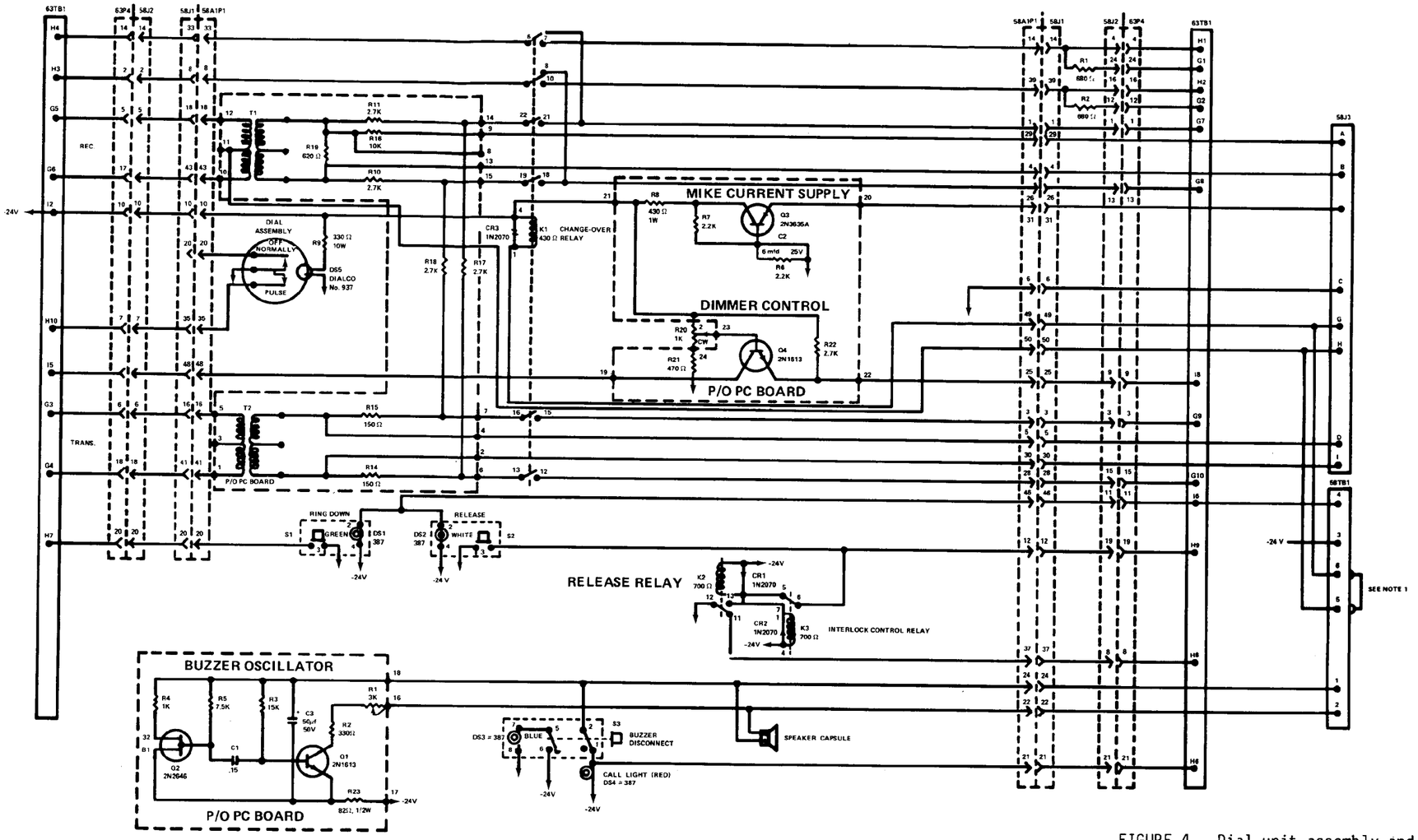


TABULATION		
ITEM	MATERIAL	FINISH
1	.188 ALUM. ALLOY	CAUSTIC ETCH & GOLD IRIDITE PAINT FRONT & SIDES GREY MIL-P-13934
2	.062 ALUM. ALLOY	CAUSTIC ETCH & GOLD IRIDITE
3	.092 ALUM. ALLOY	CAUSTIC ETCH & GOLD IRIDITE
4	.062 ALUM. ALLOY	CAUSTIC ETCH & GOLD IRIDITE
5	1 x 1 x 1/8 EXTRUDED AL ANGLE	CAUSTIC ETCH & GOLD IRIDITE
6	.062 ALUM. ALLOY	CAUSTIC ETCH & GOLD IRIDITE
7	.062 ALUM. ALLOY	CAUSTIC ETCH & GOLD IRIDITE
8	.062 ALUM. ALLOY	CAUSTIC ETCH & GOLD IRIDITE
9	.062 ALUM. ALLOY	CAUSTIC ETCH & GOLD IRIDITE
10	.062 ALUM. ALLOY	CAUSTIC ETCH & GOLD IRIDITE
11	1/2" DIA ALUM. ROD	CAUSTIC ETCH & GOLD IRIDITE

NOTE:
TOLERANCES; FRACTIONS ± 1/64; DECIMALS, ± .005

FIGURE 4. Dial unit assembly and fabrication details (sheet 2 of 6).

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NOTE: INSTALL JUMPER BETWEEN PINS 5 AND 6 ON 58TB1

FIGURE 4. Dial unit assembly and fabrication details (sheet 3 of 6).

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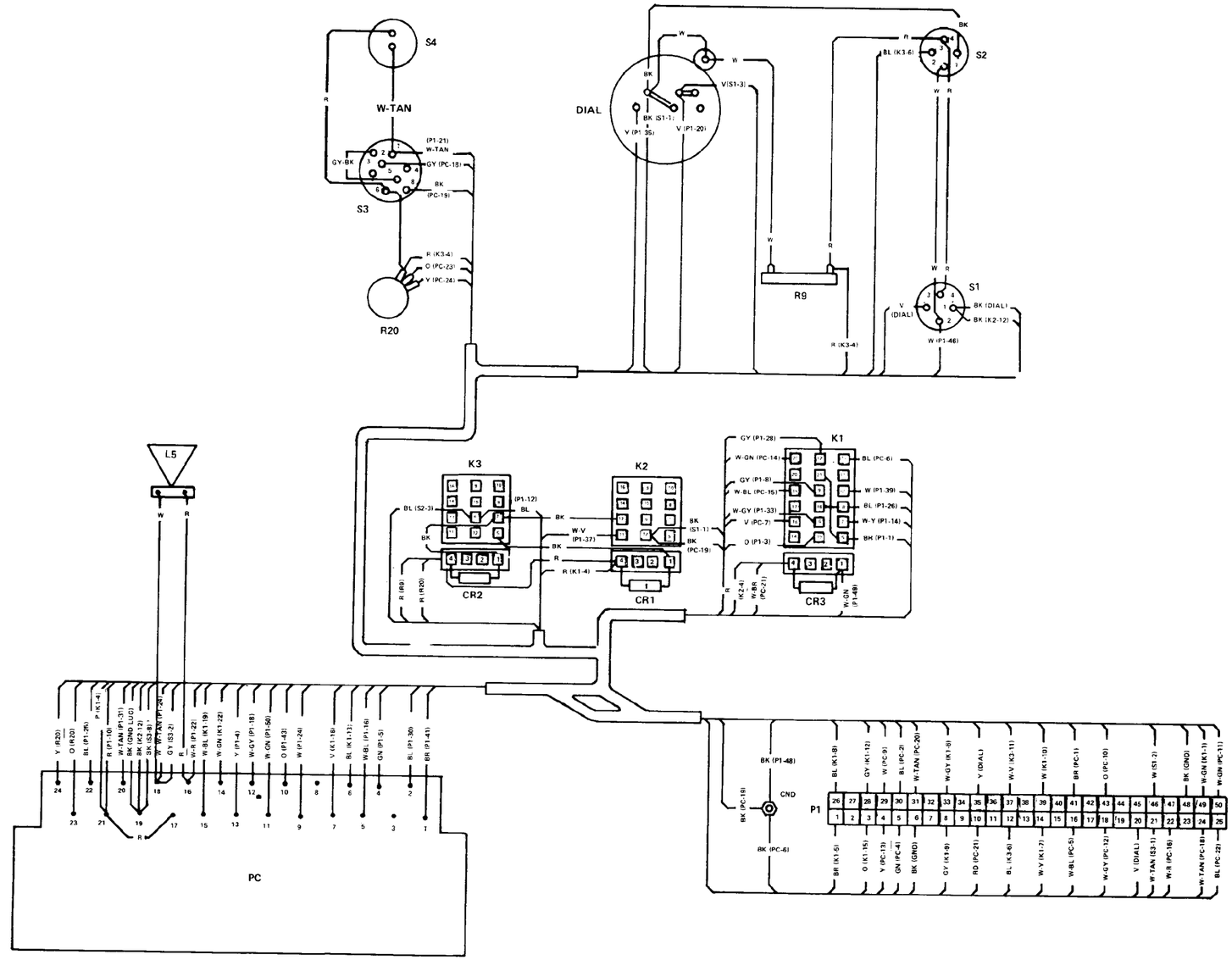


FIGURE 4. Dial unit assembly and fabrication details (sheet 4 of 6).

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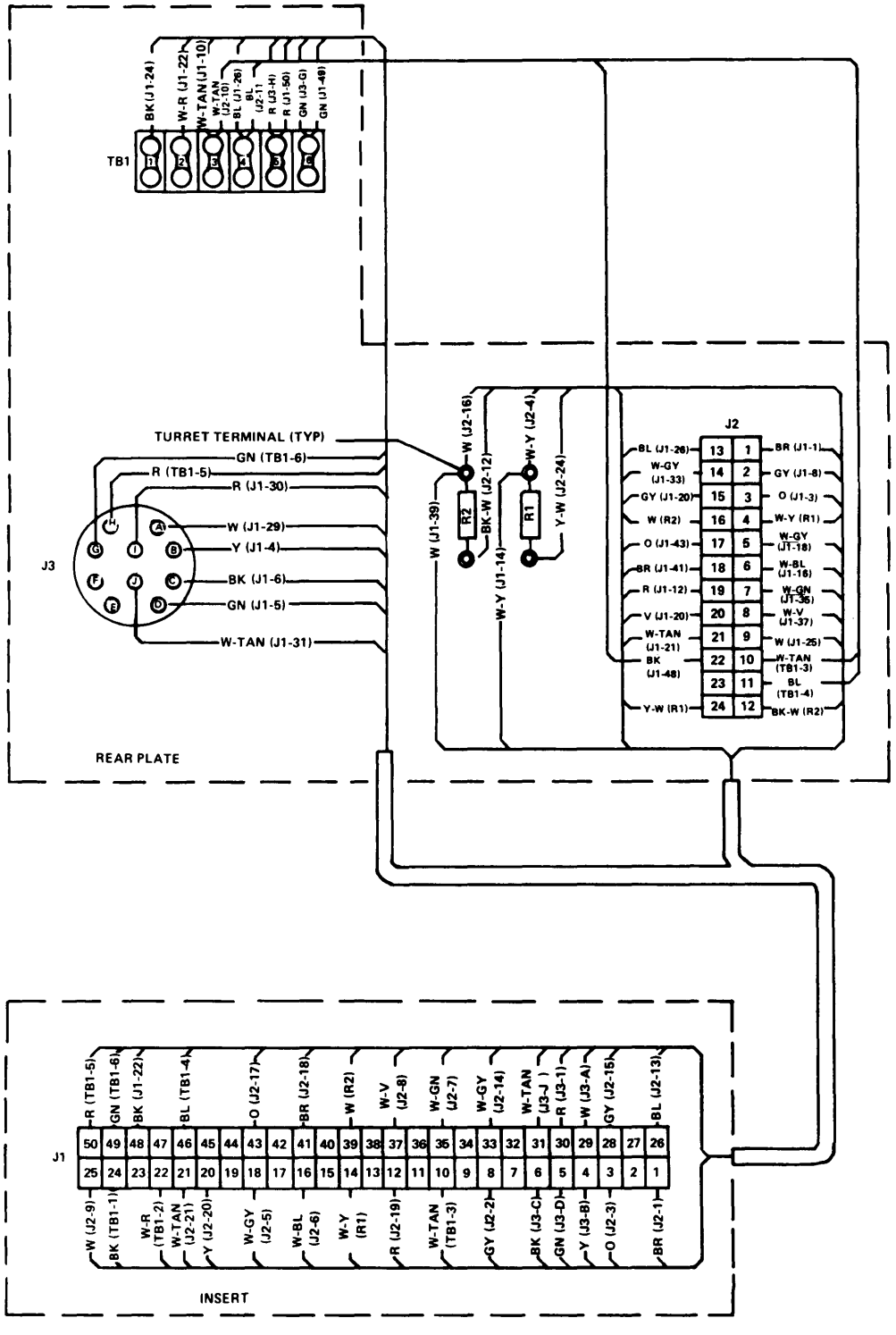
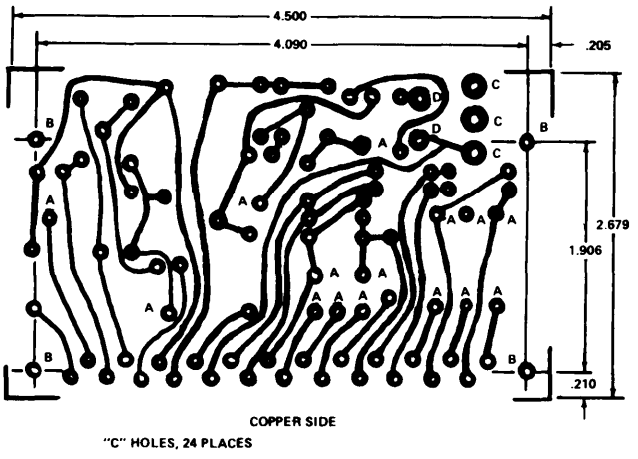


FIGURE 4. Dial unit assembly and fabrication details, insert wiring diagram (sheet 5 of 6).

NOTE: Resistors R1 AND R2 TO BE 680 OHMS, 1/4 WATT,1O PERCENT.

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LIST OF COMPONENTS

REF DESIG	DESCRIPTION	TYPE OR STOCK NUMBER
58A1C1	CAPACITOR, MONOLYTIC .15 µf, 50 wvdc	SPRAGUE ELECT. CO. TYPE SCS1
58A1C2	CAPACITOR, ELECTROLYTIC	MINIATURE, 6 µf, 25 wvdc
58A1Q1	TRANSISTOR, SILICON	MIL-S-19500/18-B, TYPE 2N1613
58A1Q2	TRANSISTOR, UNIJUNCTION	GENERAL ELECTRIC TYPE 2N2646
58A1Q3	TRANSISTOR, SILICON	FAIRCHILD CORP. SEMICONDUCTOR DIV. TYPE 2N3638A
58A1Q4	SAME AS 58A1Q1	
58A1R1	RESISTOR, VARIABLE	3000 OHMS, LINEAR TAPER MALLORY TYPE MTC 33L4
58A1R2	RESISTOR, FIXED	330 OHM, 1/4 W., 10% MIL-R-11, TYPE RC07GF
58A1R3	RESISTOR, FIXED	15000 OHM, 1/4 W., 10% MIL-R-11, TYPE RC07GF
58A1R4	RESISTOR, FIXED	1000 OHM, 1/4 W., 10% MIL-R-11, TYPE RC07GF
58A1R5	RESISTOR, FIXED	7500 OHM, 1/4 W., 10% MIL-R-11, TYPE RC07GF
58A1R6	RESISTOR, FIXED	2200 OHM, 1/4 W., 10% MIL-R-11, TYPE RC07GF
58A1R7	RESISTOR, FIXED	2200 OHM, 1/4 W., 10% MIL-R-11, TYPE RC07GF
58A1R8	RESISTOR, FIXED	430 OHM, 1W., 10% MIL-R-11, TYPE RC32GF
58A1R10	RESISTOR, FIXED	2700 OHM, 1/4 W., 10% MIL-R-11, TYPE RC07GF
58A1R11	SAME AS 58A1R10	
58A1R14	RESISTOR, FIXED	150 OHM, 1/4 W., 10% MIL-R-11, TYPE RC07GF
58A1R15	SAME AS 58A1R14	
58A1R16	RESISTOR, FIXED	10,000 1/4 10% RC07GF
58A1R17	SAME AS 58A1R10	
58A1R18	SAME AS 58A1R10	
58A1R19	RESISTOR, FIXED	620 OHM, 1/4 W., 10% MIL-R-11, TYPE RC07GF
58A1R21	RESISTOR, FIXED	470 OHM, 1/4 W., 10% MIL-R-11, TYPE RC07GF
58A1R22	SAME AS 58A1R10	
58A1T1	TRANSFORMER	600 OHM PRI, 600 SEC TRANSFORMERS, INC., TYPE T1-858
58A1T2	TRANSFORMER	120 OHM PRI, 30 OHM SEC TRANSFORMERS, INC., TYPE T1-861
1	TERMINAL, TURRET	C.T.C. MANUFACTURING CORP. TYPE X-1556-3
58A1C3	CAPACITOR, ELECTROLYTIC 50 µf, 50 wvdc	SPRAGUE ELECT. CO. TYPE 1307
58A1R23	RESISTOR, FIXED	82 OHM, 1/2 W., 10% MIL-R-11, TYPE RC20GF

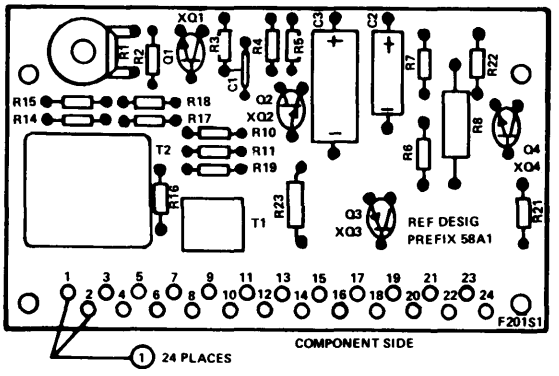
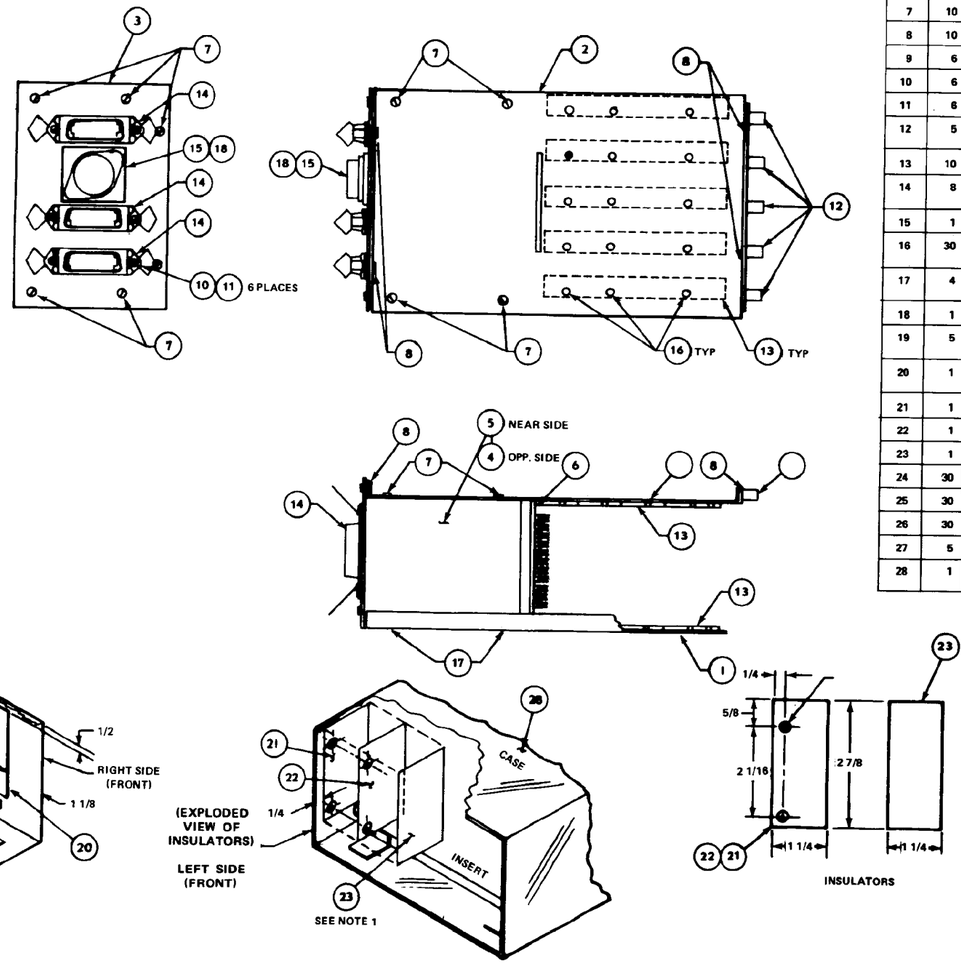


FIGURE 4. Dial unit assembly and fabrication details (sheet 6 of 6).

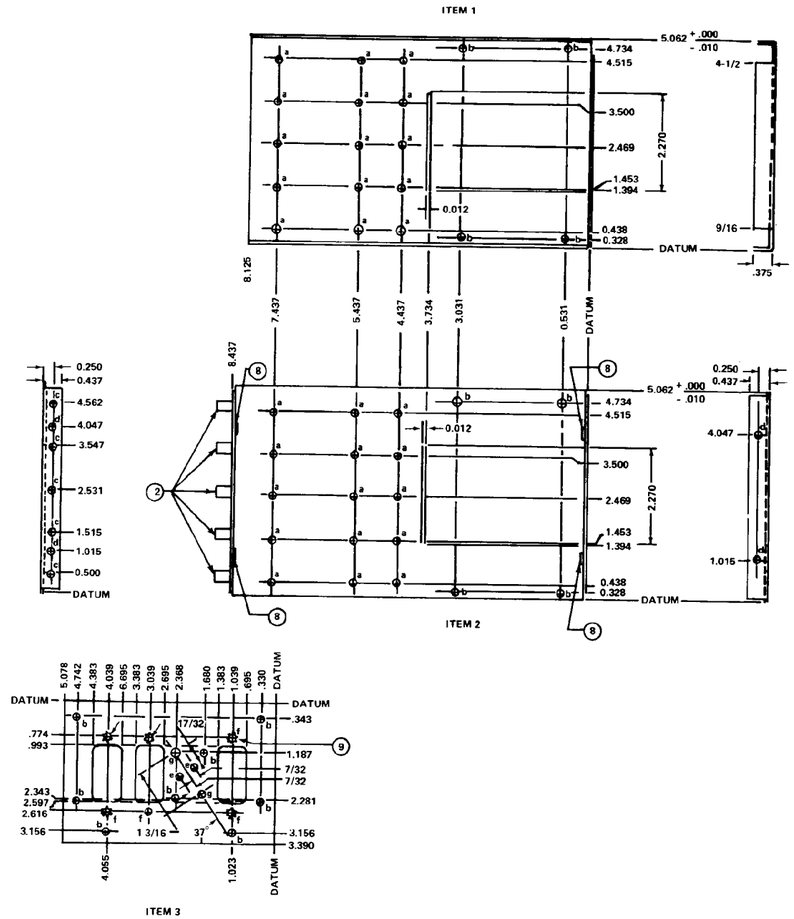
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LIST OF MATERIAL				
ITEM	QTY	DESCRIPTION	TYPE OR STOCK NO.	REMARKS
1	1	TOP PLATE		
2	1	BOTTOM PLATE		
3	1	BACK PLATE		
4	1	SIDE PLATE		
5	1	SIDE PLATE		
6	1	MASTER BOARD		SEE DWG 41031
7	10	SCREW, PAN HEAD	NO. 4-40 x 1/4 CAD. PLATE	
8	10	PEM FASTENER	NO. 4-40 THD.	
9	6	PEM FASTENER	NO. 3-48 THD.	
10	6	SCREW PAN HEAD	NO. 3-48 x 5/8 CAD. PLATE	
11	6	WASHER, LOCK	NO. 3 CAD. PLATE	
12	5	STANDOFF	1/4 DIA x 3/8 LONG NO. 8-32 CAD. PLATE	
13	10	GUIDE, PRINTED CIRCUIT		
14	8	CONNECTOR, CHASSIS TO CABLE	AMPHENOL NO. 57-40240	
15	1	TRANSISTOR	2N3055 (RCA)	
16	30	EYELET	.067 D. x 1/8 LG .145 D. HEAD	
17	4	EYELET	.123 D. x 5/32 LG .197 D. HEAD	
18	1	SOCKET, TRANSISTOR	MOTOROLA TYPE MK 15	
19	5	SCREW, CAPTIVE	8-32 x 15/16 SOUTHCO 54-37-809-24	
20	1	NAMEPLATE	FABRICATE	SEE RHA-6-F-41080
21	1	INSULATOR, VINYL	5 MIL, ADHESIVE BACK	
22	1	INSULATOR, MYLAR	2 MIL, ADHESIVE BACK	
23	1	INSULATOR, MYLAR	2 MIL, ADHESIVE BACK	
24	30	SCREW	6-32 x 1/4 NICKEL PLATE	
25	30	CONTACT, LAMP	FABRICATE	
26	30	INSERT, THREADED	6-32 UNC - 1/4	
27	5	PLATE	FABRICATE	
28	1	CASE	FABRICATE	SEE RHA-6-41022

FIGURE 5. Line select unit insert (sheet 1 of 6)).

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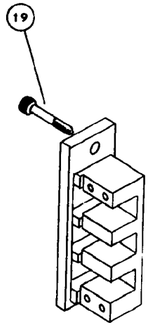


NOTES:

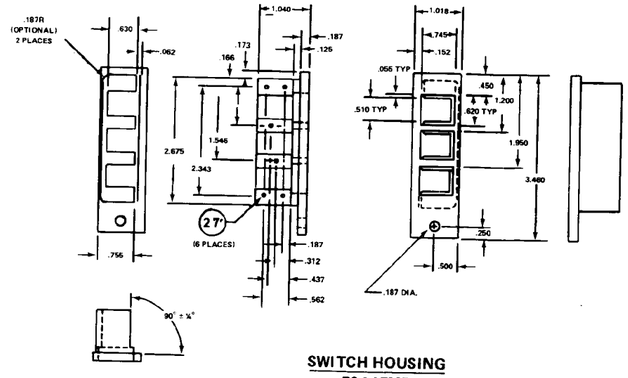
1. HOLE SIZES:
   a. 3/32 DIA.
   b. 1/8 DIA.
   c. NO. 12 DRILL
   d. NO. 19 DRILL FOR NO. 4-40 PEM FASTENER
   e. 5/32 DIA. PEM FASTENER
   f. 11/64 DIA. FOR 5-48 PEM FASTENER
   g. 3/16 DIA.
2. MATERIAL:
   ITEM 1:0.031 ALUM ALLOY
   ITEMS 2, 3, 4, 5:0.062 ALUM ALLOY
3. FINISH:
   ITEMS 1, 2, 3, 4, 5, ETCH AND GOLD IRIDITE
4. 1 ITEM NO. (SEE SHEET 1)
5. TOLERANCES:FRACTIONS, ± 1/64; DECIMALS, ± 0.005

FIGURE 5. Line select unit insert, fabrication details (sheet 2 of 6).

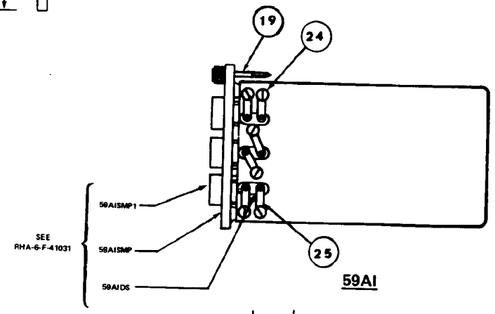
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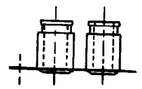
59AISIMP4

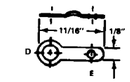


SWITCH HOUSING
59A15MP

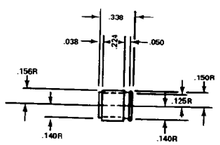


59AI

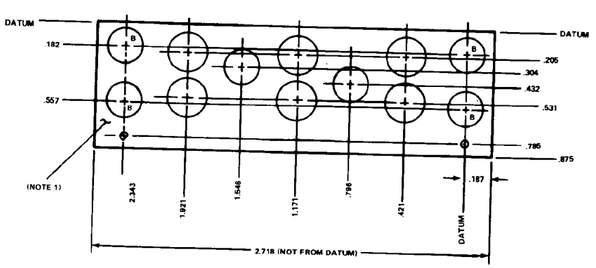




ITEM 25



P/O ITEM 27



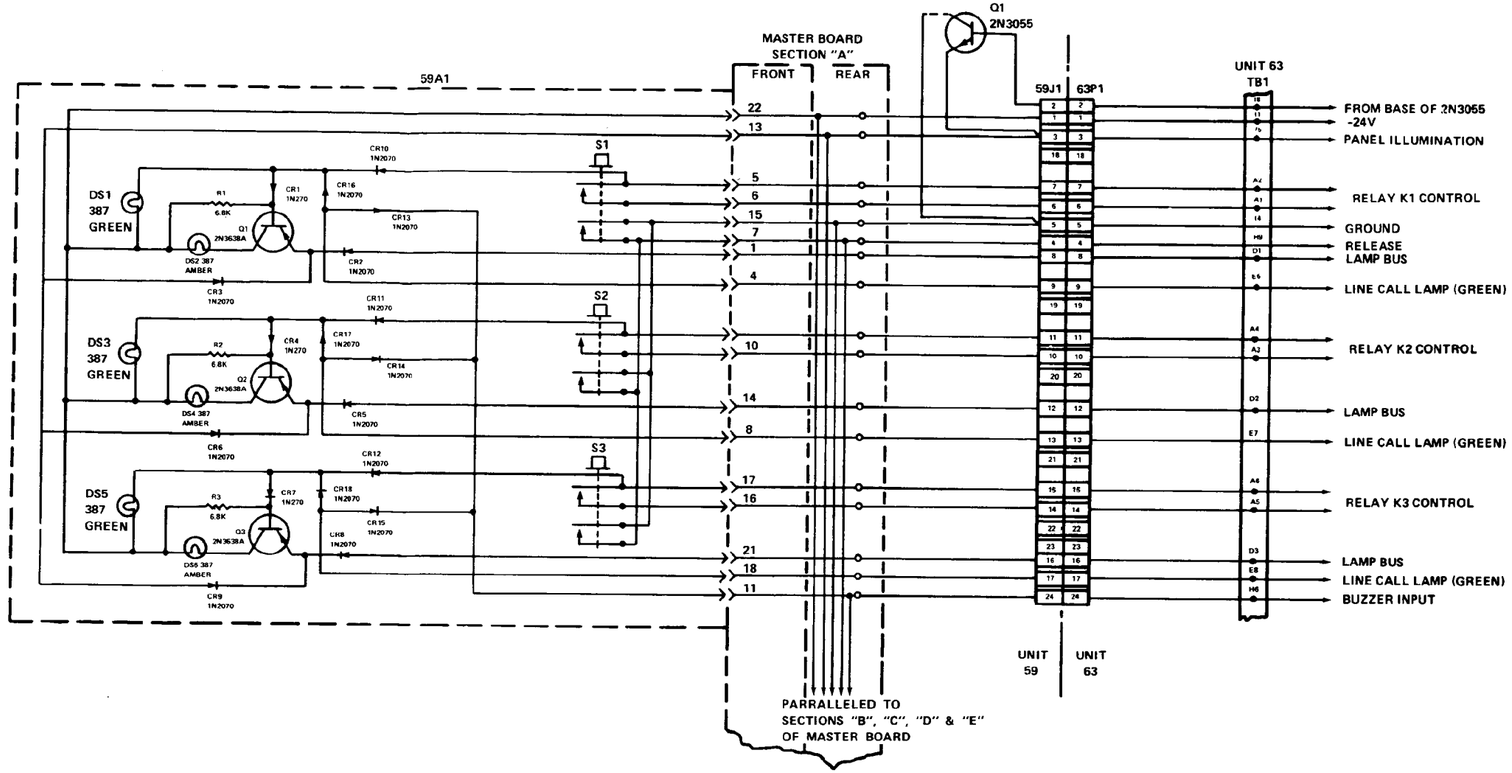
ITEM 27

NOTES:

1. HOLE SIZES:
   a. DRILL 0.290 DIA., – 0.005 + 0.000
   b. DRILL 0.250 DIA.
   c. DRILL 1/16 INCH DIA.
   d. DRILL NO. 27
   e. DRILL NO. 48
2. MATERIAL:
   ITEM 25 – PHOSPHOR BRONZE
   ITEM 27 – HALF HARD BRASS PLATE 0.018 THK
   FREE CUTTING BRASS TUBE, O.D. NO. B12,
   I.D. NO. 0.250
3. FINISH:
   ITEM 27 – NICKEL PLATE AFTER ASSEMBLY
4. TOLERANCE ± 0.005 UNLESS OTHERWISE SHOWN

FIGURE 5. Line select unit insert (sheet 3 of 6).

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NOTES:

1. RESISTORS ARE 1/4 WATT, 10 PERCENT
2. 1N2070 DIODES ARE TO BE SUBMINIATURE TYPE (ELECTRON RESEARCH, INC., OR EQUIVALENT)
3. TRANSISTORS ON PC BOARD ARE TO BE PROVIDED WITH ELCO TRANSISTOR SOCKETS TYPE 05-3305
4. THE 2N3638A TRANSISTOR SPECIFIED IS TO HAVE A TO-5 PIN CONFIGURATION SIMILAR TO FAIRCHILD'S

FIGURE 5. Line select unit insert, schematic diagram (sheet 4 of 6).

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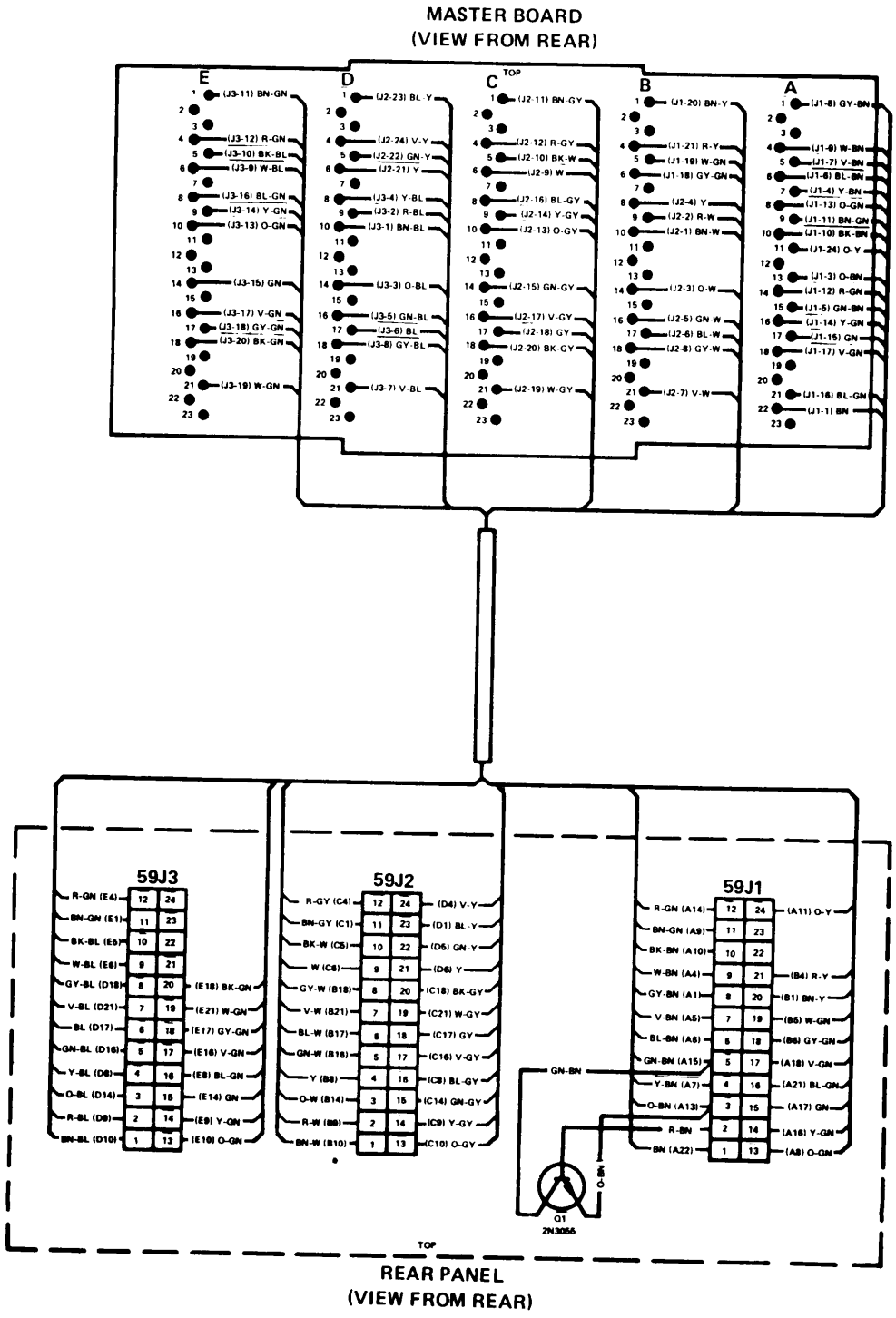
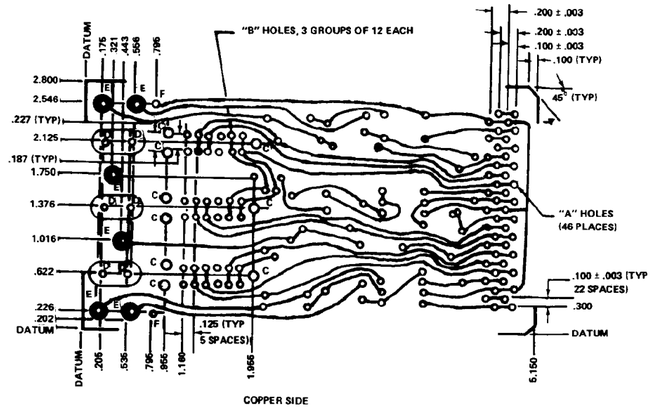
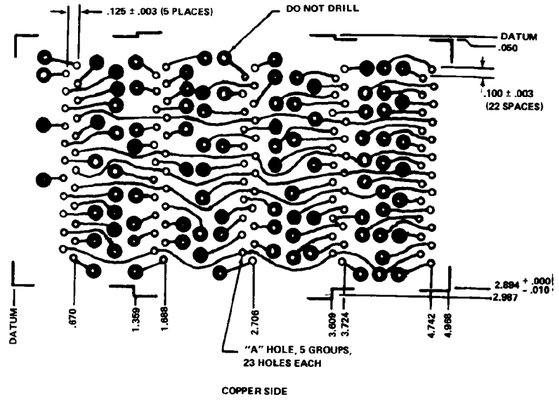


FIGURE 5. Line select unit insert (sheet 5 of 6).

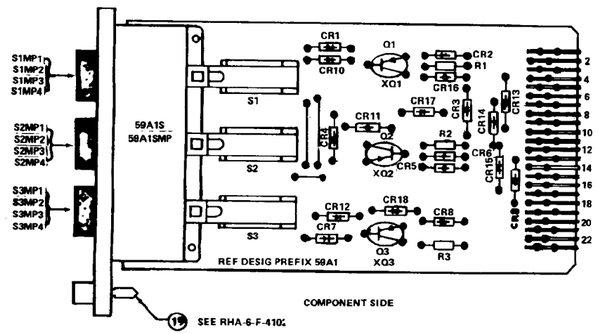
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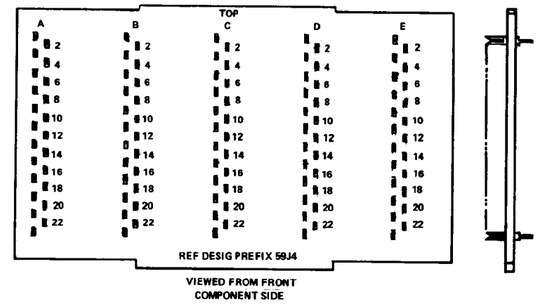


COPPER SIDE



COPPER SIDE





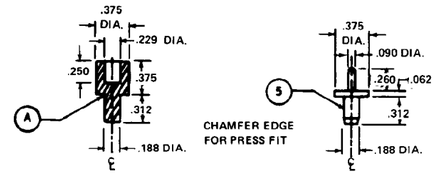
LIST OF COMPONENTS		
REF DESIG	DESCRIPTION	TYPE OR STOCK NUMBER
59J4	CONNECTOR ASSEMBLY	
59J4A THRU 59J4E	CONNECTOR STRIP	02-023-12-1-200
59A1	LINE SELECT MODULE	
59A1CR1	SEMICONDUCTOR DEVICE, DIODE	MIL-S-19500/256 / TYPE 1N270
59A1CR2	SEMICONDUCTOR DEVICE, DIODE	ERIE TECHNOLOGICAL PRODUCTS TYPE ED3004A/1N2070
59A1CR3	SAME AS CR2	
59A1CR4	SAME AS CR1	
59A1CR5	SAME AS CR2	
59A1CR6	SAME AS CR2	
59A1CR7	SAME AS CR1	
59A1CR8	SAME AS CR2	
59A1CR9	SAME AS CR2	
59A1CR10	SAME AS CR2	
59A1CR11	SAME AS CR2	
59A1CR12	SAME AS CR2	
59A1CR13	SAME AS CR2	
59A1CR14	SAME AS CR2	
59A1CR15	SAME AS CR2	
59A1CR16	SAME AS CR2	
59A1CR17	SAME AS CR2	
59A1CR18	SAME AS CR2	
59A1DS1	LAMP, INDICATING PART OF 59A1S1	TYPE NO. 387 (GREEN)
59A1DS2	LAMP, INDICATING PART OF 59A1S1	TYPE NO. 387 (AMBER)
59A1DS3	SAME AS DS1 PART OF 59A1S2	
59A1DS4	SAME AS DS2 PART OF 59A1S2	
59A1DS5	SAME AS DS1 PART OF 59A1S3	
59A1DS6	SAME AS DS2 PART OF 59A1S3	
59A1Q1	TRANSISTOR, SILICON	FAIRCHILD CORP., SEMICONDUCTOR DIV. TYPE 2N3638A
59A1Q2	SAME AS Q1	
59A1Q3	SAME AS Q1	
59A1R1	RESISTOR, FIXED	6300 OHM, 1/4 W., 10%
59A1R2	SAME AS R1	
59A1R3	SAME AS R1	
59A1S	SWITCH, ASS., ILLUM. PLUG IN	3 SECTIONS, 4 "C" GROUPS EACH VELCO SYSTEMS TYPE VSA03
59A1S1	SWITCH, ILLUM., 1 SECTION, 4 "C" GROUPS	PART OF 59A1S
59A1S2	SAME AS S1	
59A1S3	SAME AS S1	
59A1SMP	HOUSING, SWITCH, GRAY	VELCO SYSTEMS TYPE VSM01-8
59A1S1MP1	PUSH BUTTON	VELCO SYSTEMS TYPE VMP03
59A1S1MP2	FILTER, WHITE	VELCO SYSTEMS TYPE VMP04-8
59A1S1MP3	DIFFUSER	VELCO SYSTEMS TYPE VMP02
59A1S1MP4	REFLECTOR	VELCO SYSTEMS TYPE VMP01
59A1S2MP1	SAME AS S1MP1	
59A1S2MP2	SAME AS S1MP2	
59A1S2MP3	SAME AS S1MP3	
59A1S2MP4	SAME AS S1MP4	
59A1S3MP1	SAME AS S1MP1	
59A1S3MP2	SAME AS S1MP2	
59A1S3MP3	SAME AS S1MP3	
59A1S3MP4	SAME AS S1MP4	
59A1XQ1	SOCKET, TRANSISTOR	ELCO CORP. TYPE 05-3305
59A1XQ2	SAME AS XQ1	
59A1XQ3	SAME AS XQ1	

NOTES:

1. HOLE SIZES:
   a. NO. 55 DRILL
   b. NO. 52 DRILL
   c. NO. 49 DRILL
   d. 21/64 IN. DIA. MAKE CUTOUT AS SHOWN
   e. NO. 27 DRILL
   f. 1/16 DIA.
   g. ALL OTHER HOLES TO BE NO. 56 DRILL
2. MATERIAL:
   59A1-0.062 THK TYPE GE GLASS EPOXY, 2-OZ. COPPER FOIL ONE SIDE IAW MIL-P-13949
   59J4-0.093 THK TYPE GE GLASS EPOXY, 2-OZ. COPPER FOIL ONE SIDE IAW MIL-P-13949
3. FIN
   59-LEAD PLATE IAW MIL-STD-275
4. COMPONENT SIDE OF BOARDS TO BE SILK-SCREENED TO SHOW COMPONENTS AND THEIR DESIGNATIONS
5. TOLERANCE: ±0.005 UNLESS OTHERWISE SHOWN

FIGURE 5. Line select unit insert (sheet 6 of 6).

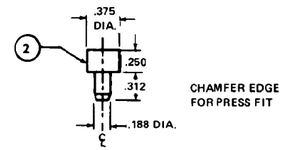
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DETAIL "A"

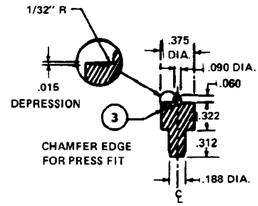
DETAIL "B"

RIVET EXTRACTORS
SCALE: 2:1



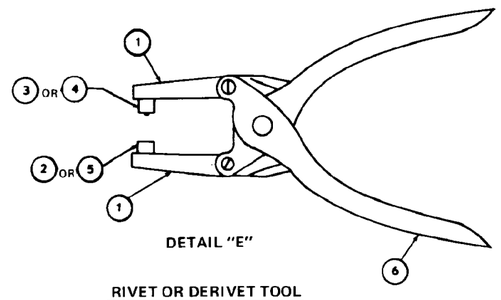
DETAIL "C"

CHAMFER EDGE
FOR PRESS FIT



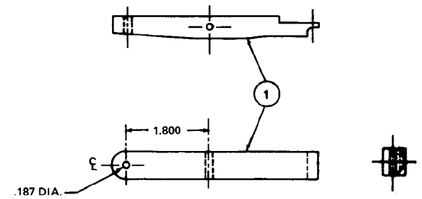
DETAIL "D"

RIVET HEADS
SCALE: 2:1



DETAIL "E"

RIVET OR DERIVET TOOL
NO SCALE



DETAIL "F"

BODY MODIFICATION
SCALE; 1:1

CONTRACT FURNISHED MATERIAL				
ITEM NO.	QTY	DESCRIPTION	NAVY OR COML DESIGNATION	REMARKS
7	200	RIVETS, HOLLOW END ALUM.		1/8" x 3/16"
6	2	HANDLE ASSY	SARGENT & CO. NEW HAVEN, CONN.	SEE DETAIL "E"
5	1	RIVET EXTRACTOR	FABRICATE	SEE DETAIL "B"
4	1	RIVET EXTRACTOR	FABRICATE	SEE DETAIL "A"
3	1	RIVET HEAD	FABRICATE	SEE DETAIL "D"
2	1	RIVET HEAD	FABRICATE	SEE DETAIL "C"
1	2	SCDY	FABRICATE	SEE DETAIL "F"

FIGURE 6. Rivet tool set.

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